Practical Assignment – I

Getting Familiarity with C++

1. Write a C++ program to print Hello World.
2. Write a C++ program to display any number entered by the user.
3. Write a C++ program to input three numbers and display its total and average.
4. Write a C++ program to enter Day Temperature of 5 cities of Gujarat. Display average temperature.
5. Write a C++ program to find simple interest.
   SI = (P*R*N)/100 with float values
6. Write a C++ program to calculate area of circle.
   Area = $\pi * r * r$
7. Write a C++ program to read two numbers from user and print their addition, subtraction, multiplication and division on screen.
8. Write a C++ program to input marks of five subjects of a student and calculate total marks and percentage.

ASSIGNMENT -II

Getting familiar with C++

Class and Object

1. WAP to print your name, age and city and pin code on screen (Using Class).
2. Write a program to print the area of a rectangle by creating a class named 'Area' having two functions. First function named as 'setDim' takes the length and breadth of the rectangle as parameters and the second function named as 'getArea' returns the area of the rectangle. Length and breadth of the rectangle are entered through keyboard.
3. WAP to display addition, subtraction, multiplication and division of two integers on screen.

   Declare Class Calculation
   Declare data member num1 and num2 in Private section
   Write member function for initialize num1 and num2
   Write member function for each operation.

4. WAP to find area of circle. Area of Circle = PI * r * r Where, PI = 3.14
   (Using Class and Object)

5. Write a C++ program to create a class for student to get and print details of a student. Following are the details of a student:

   Studid, name ,sem, branch

6. Write a C++ program to demonstrate ATM money withdrawal process by taking following private data members:

   Accountno, balance;

   The withdrawal function should return remaining balance to the user and should deduct withdrawal amount from balance. If withdrawal amount > balance print appropriate message on screen (Not enough balance)

   The Deposit function should return updated balance to user.

7. Write a C++ program to create a class for student to get and print details of a student. Following are the details of a student:

   Student_id, Name, Branch,Sub1_mark,Sub2_mark,Sub3_mark,Sub4_mark,Sub5_mark

   Write member function to calculate Percentage, Class (Dist,First,Second,Pass) of student

8. C++ program to create a class for Employee to get and display following Employee information:

   Empcode, Basicsalary

   Write member function to calculate Net salary
   DA=174% of Basic salary
   HRA=10% of Basic salary
   TA=500
   Income tax=5 % of basic salary, if basic salary >50000
   Income tax=0, if Basic salary <=50000

   Netsalary=(basicsalary+da+hra+ta) - income tax

9. Do above program for 5 number of Employees.(Using Array of Object)

10. Write a program to read time in hh:mm:ss and display answer in only seconds. For example if user enters 2:15:30 then it should display 8130 seconds.
    Input:
    Enter Hours:2
    Enter minutes:15
    Enter Seconds:30
    Output:
    8130 seconds

ASSIGNMENT -III
Getting familiar with C++
Constructor

1. Write a program to print the area of a rectangle by creating a class named 'Area' having one function. Length and breadth of the rectangle are entered through keyboard using Parameterized constructor.

2. WAP to display addition, subtraction, multiplication and division of two integers on screen.

   Declare Class Calculation

   Write Parameterized constructor for initialize num1 and num2

   Write member function for each operation.

3. Write a C++ program to demonstrate ATM money withdrawal and deposit process by taking following private data members:

   Accountno, balance;

   Account no and balance data member initialize using parameterized constructor

   Write three function 1. Deposit 2. Withdraw 3. Balance

   Write menu driven choice

   1. Deposit
   2. Withdraw
   3. Balance
   4. Exit

   Program stop execution when user enter choice 4.

4. Create a class “Mobile” with attributes: brand, price, color, width, height. Use constructor to set default values of these attributes. Write function to display details of all attributes.

5. Create a class “Mobile” with attributes: brand, price, color. Enter detail of five different mobile. (Using Array of object).

Display total number of mobile having price greater than 5000.

Display Brand, Price and color for all mobiles for price range 1000 to 10000

6. C++ program to create a class for Employee to get and display following Employee information:

Empcode,Emp name, Basicsalary

Count the created objects using static member.

7. Create a class "Student" having following data members:

studid, name, marks (of 5 subject), percentage.

Use getdata and show functions to input and display student data.
Create private function to calculate percentage.
Display detail of student who secured highest percentage.
Create N number of student using array of object.

8. Create a class Student with Rollno,mark1,mark2,mark3 data member and write appropriate function to setdata .Write another class name sport with sport_mark data member and set sport_mark. Find Resul(mark1+mark2+mark3+sport_mark) of student using Friend function

ASSIGNMENT IV
Getting familiar with Inheritance in C++
Inheritance

1. Write a program that defines a shape class with a constructor that gives value to width and height. Then define two sub-classes triangle and rectangle, that calculate the area of the shape. In the main, define two objects a triangle and a rectangle and then call the area () function.

2. Write a program with a mother class animal. Inside it define a name and an age variables, and set_value () function. Then create two sub class Zebra and Dolphin which write a message telling the age and name of animal, also giving some extra information for both sub class (e.g. place of origin).place of origin of zebra is Earth and place of origin of dolphin is water.

3. Write a program as per following details

Create one base class Teacher

Data members Name, Department, College name, Email id.

Create sub classes for Math Teacher, English Teacher, and Science Teacher

Data member Qualification, Expertise and salary.

Display following details for each teacher

Name:

Department:

College name:

Email id:

Qualification:

Expertise:

Salary:

4. Write a program as per following details

Create one base class DRUG with following data members

Category- (i.e. **stimulants, inhalants, cannabinoids)**

Date_of_manufacture, Company name

Create one sub class TABLET derived from DRUG with following data members

Tablet name, Price

Create one sub class PainReliever derived from TABLET with data member

Dosage_units: i.e(1 or 2 or 3)
Side_effects : i.e (Nausea, Drowsiness, Dizziness.)

Use_within_days: i.e(10 or 20 or 30).

Use appropriate member function for setting and Getting above details and display details in main function.

5. Write a program as per following details

Create one base class HOTEL with following data members

Hotel_name,

Hotel_type i.e(Three star,five star)

City

Hotel_rate i.e(2000,3000,5000)

Create one base class FLIGHT with following data members

Flight_no

Source city

Destination city

Seat no

Create one sub class PASSENGER derived from HOTEL and FLIGHT with following data members

Name, Age, city

Write appropriate member functions in each class and display all information in main

6. Write a program as per following details

Create one base class PERSON with following data members

Name, College name

Create one sub class STUDENT derived from PERSON with following data members

Student_id , Marks of five subject, percentage

Member function:

showResult( )-Calculate total,percentage and finding class(Dist,First,second,pass)

Create one sub class EMPLOYEE derived from PERSON with following data members

Emp_id, qualification , basic salary

Member function to calculate Net salary and print Net salary

DA=189% of Basic salary
HRA=10% of Basic salary
TA=500
Income tax=5 % of basic salary, if basic salary >50000
Income tax=0, if Basic salary <=50000

Netsalary=(basicsalary+da+hra+ta) - income tax

Write appropriate setter function in each class and display detail of student and employee in main.

ASSIGNMENT V
Getting familiar with Operator Overloading

1. Write a program to overload unary minus operator

   Ex. Input : obj(10,-20,30) Output : -10,20,-30 (-obj)

2. Write a program to overload pre decrement operator (obj1= --obj)

3. Write a program to overload post decrement operator (obj1=obj--)

4. Write a program to overload Binary *(multiply) operator for object of same class

Ex. class student s1,s2,s3 s3=s1*s2

5. Write a program to overload Binary * (multiply) operator for object of different class

Ex. Class unit_test, class Practical Practical result=object of unit_test+object of practical

6. Write program to overload = = operator to compare two object of student class

```
Class student
{
        Int mark;
        Public:
        Student (int tmark)
        {
         mark=tmark;
        }
};

Int main() {

Student s1(100),student s2(200),student s3(100);

If(s1==s2)

   Cout<<" no same marks"

If(s1==s3)

  Cout<< "same marks"
```

ASSIGNMENT VI
Working with File

1.Write a program to create a file “Product.txt” and write data as Product Name, Product Price and Batch Number in a file.

2. Write a program to append Product Type and Product Weight in a file “Product.txt”.

3. Write a program to create a file “Patient.txt” and enter Patient ID, Patient Name, Disease Name and City in a file. Now read patient details and display all the details on a terminal.

4. Write a program to append Doctor Name and Doctor Speciality in “Patient.txt” file and display the data on a console.

5.Create a class “Mobile” with attributes: brand, price, color. Use getData() values of these attributes and store object in file. Write function displayData() to display object from file.

6. Create a class “Student” having following data members:

studid,name,marks (of N subject), percentage. Create private function to calculate percentage. Use getData() functions get data from keyboard and store data to the file. Write function displayData() to display object from file.

Getting familiar with basic SQL Syntax (DDL, DML Statements)

1. Create following table which has following structure:

a. **TABLE NAME: EMP**

Field Name	Data Type	Null
EMPNO	NUMBER(4) PK	
ENAME	VARCHAR2(20)	
JOB	VARCHAR2(20)	
MGR	NUMBER(4)	YES
HIREDATE	DATE	
SAL	NUMBER(7,2)	
COMM	NUMBER(7,2)	YES
DEPTNO	NUMBER(2) FK	

TABLE NAME : DEPT

Field Name	Data Type	Null
DEPTNO PK	NUMBER(2)	
DNAME	VARCHAR2(20)	
LOC	VARCHAR2(20)	

b. **TABLE NAME : SALGRADE**

Field Name	Data Type	Null
GRADE	NUMBER(2)	
LOSAL	NUMBER (4)	
HISAL	NUMBER (4)	

2. Insert following records in respective tables.

Table Name: EMP

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7369	SMITH	CLERK	7902	17-DEC-80	800		20
7499	ALLEN	SALESMAN	7698	20-FEB-81	1600	300	30
7521	WARD	SALESMAN	7698	22-FEB-81	1250	500	30
7566	JONES	MANAGER	7839	02-APR-81	2975		20
7654	MARTIN	SALESMAN	7698	28-SEP-81	1250	1400	30
7698	BLAKE	MANAGER	7839	01-MAY-81	2850		30
7782	CLARK	MANAGER	7839	09-JUN-81	2450		10
7788	SCOTT	ANALYST	7566	09-DEC-81	3000		20
7839	KING	PRESIDENT		17-NOV-81	5000		10
7844	TURNER	SALESMAN	7698	08-SEP-81	1500	0	30
7876	ADAMS	CLERK	7788	12-JAN-83	1100		20
7900	JAMES	CLERK	7698	03-DEC-81	950		30
7802	FORD	ANALYST	7566	03-DEC-81	3000		20
7934	MILLER	CLERK	7782	23-JAN-82	1300		10

Table Name: DEPT

DEPTNO	DNAME	LOC
10	ACCOUNTING	NEW YORK
20	RESEARCH	DALLA
30	SALES	CHICAGO
40	OPERATIONS	BOSTON

Table Name: SALGRADE

GRADE	HISAL	LOSAL
1	700	1200
2	1201	1400
3	1401	2000
4	2001	3000
5	3001	9999

3. Solve the following queries.

1.	Find out the name of all employees.
2.	Display all information of employee.
3.	Display all information of employee whose job type is “SALESMAN”.
4.	Display all information of employee whose Dept No is 20.
5.	Display employee information whose hire date is in month of DEC.
6.	Display employee information whose salary is greater than 3000.
7.	Find the names of all employees having ‘A’ as the second letter in their names.
8.	Display department information.
9.	Display Salary Grade Information.
10.	Display Name, Job of employee whose manager name is “BLAKE”.
11.	Display Name, Salary of employee whose commission is null.
12.	Display Name, Salary, Hiredate of employee whose Hiredate is in the year of 81.
13.	Update salary by 5% whose department no is 10.
14.	Update commission comm=sal*5% whose job type is “MANAGER”.
15.	Display employee name whose name start with ‘A’.
16.	Display employee details that have maximum salary.
17.	Display employee details that have minimum salary.
18.	Display average salary department wise.
19.	Display employee details whose salary is between 1000 to 2000.
20.	Display employee information department wise.

21.	Display your name in upper, lower and title case.
22.	Display all Employee names and their length after trimming them
23.	Extract first 3 characters from your name and display the same.
24.	Extract last 4 characters from your name and display the same.
25.	Extract only 3 to 7 characters of your full name.
26.	Find out the absolute values of the following: -20, 30, 25, -400, -13, -11, 34, -99.
27.	Update the employee table so each employee name will contain XXXXX at the end of the same.
28.	Now display the same such that they won't display XXXXX at the end of each name.
29.	Round the following numbers by 5, 2, 1, 3, 4 decimal places: 4.091978,232.837733, 822.2325443.

Getting familiar with basic SQL Syntax (DDL, DML Statements)

2. Create following table which has following structure:

a. TABLE NAME: CLIENT_MASTER

Field Name	Data Type	Constraint
CLIENT_NO	VARCHAR2(6)	Primary Key /First letter must Start with 'C'.
NAME	VARCHAR2(20)	Not null
ADDRESS1	VARCHAR2(30)	
ADDRESS2	VARCHAR2 (30)	
CITY	VARCHAR2(15)	
STATE	VARCHAR2 (15)	
PINCODE	NUMBER(6)	
BAL_DUE	NUMBER(10,2)	

a. TABLE NAME : PRODUCT_MASTER

Field Name	Data Type	Constraint
PRODUCT_NO	varchar2(6)	Primary key Must start with P
DESCRIPTION	VARCHAR2(14)	NOT NULL
PROFIT_PERCENT	NUMBER(4,2)	NOT NULL
UNIT_MEASURE	VARCHAR2(10)	NOT NULL
QTY_ON_HAND	NUMBER(8)	NOT NULL
REORDER_LVL	NUMBER(8)	NOT NULL
SELL_PRICE	NUMBER(8,2)	NOT NULL, Cannot be 0
COST_PRICE	NUMBER(8,2)	NOT NULL, Cannot be 0

a. TABLE NAME : SALESMAN_MASTER

Field Name	Data Type	Null
SALESMAN_NO	VARCHAR2(6)	Primary Key / first letter must start With ‘S’
SALESMAN_NAME	VARCHAR2(20)	NOT NULL
ADDRESS1	VARCHAR2(30)	NOT NULL
ADDRESS2	VARCHAR2(30)	
CITY	VARCHAR2(20)	
PINCODE	VARCHAR2(6)	
STATE	VARCHAR2(20)	
SAL_AMT	NUMBER(8,2)	Not Null,cannot be 0
TGT_TO_GET	NUMBER(6,2)	Not Null,cannot be 0
YTD_SALES	NUMBER(6,2)	NOT NULL
REMARKS	VARCHAR2(60)	

Table Name :sales_order

Field Name	Data Type	Null
S_ORDER_NO	VARCHAR2(6)	Primary Key / first letter must start With ‘O’
S_ORDER_DATE	DATE	
CLIENT_NO	VARCHAR2(6)	Foreign Key references client_no Of client_master table
DELY_ADDR	VARCHAR2(25)	
SALESMAN_NO	VARCHAR2(6)	Foreign Key references salesman_no Of salesman_master table.
DELY_TYPE	CHAR(1)	delivery : part(P)/full(F),Default ‘F’
BILLED_YN	CHAR(1)	
DELY_DATE	DATE	cannot be less than s_order_date
ORDER_STATUS	VARCHAR2(10)	values (‘In Process’,’Fulfilled’, ‘BackOrder’,’Canceled’)

Perform Following Changes on Product_Master table

1) Change Datatype of Product_no field from number(2) to varchar2(6).
2) Assign primary key to product_no field
3) Add Constraint to product_no field (Must start with P)

Table Name : sales_order_details

Field Name	Data Type	Null
S_ORDER_NO	VARCHAR2(6)	Foreign Key references S_order_no of sales_order table.
PRODUCT_NO	VARCHAR2(6)	Foreign Key references Product_no of product_master table.
QTY_ORDERED	NUMBER(8)	
QTY_DISP	NUMBER(8)	
PRODUCT_RATE	NUMBER(10,2)	

Table Name :challan_header

Field Name	Data Type	Null
CHALLAN_NO	VARCHAR2(6)	Primary Key / first two letters must Start with 'CH'.
S_ORDER_NO	VARCHAR2(6)	Foreign Key references s_order_no of Sales_order table.
CHALLAN_DATE	DATE	NOT NULL
BILLED_YN	CHAR(1)	values('Y','N'),Default 'N'.

Table Name : Challan_Details

Field Name	Data Type	Null
CHALLAN_NO	VARCHAR2(6)	Foreign Key references Challan_no of challan_header table.
PRODUCT_NO	VARCHAR2(6)	Foreign Key references Product_no of product_master table.
QTY_DISP	NUMBER(4,2)	NOT NULL

1. **. Insert following records in respective tables.**

Table Name: CLIENT_MASTER

CLIENT NO.	CLIENT NAME	CITY	PINCODE	STATE	BAL_DUE
C00001	Ivan Bayross	Bombay	400054	Maharashtra	15000
C00002	Vandana Saitwal	Madras	780001	Tamil Nadu	0
C00003	Pramada Jaguste	Bombay	400057	Maharashtra	5000
C00004	Basu Navindgi	Bombay	400056	Maharashtra	0
C00005	Ravi Shreedharan	Delhi	100001		2000
C00006	Rukmini	Bombay	400050	Maharashtra	0

Table Name: PRODUCT_MASTER

PRODUCT no	DESCRIPTION	PROFIT_PERCENT	UNIT_MEASU	QTY_ON_HAND	REORDER_LVL	SELL_PRICE	COST_PRICE
P00001	1.44 Floppies	5	Piece	100	20	525	500
P03453	Monitors	6	Piece	10	3	12000	11280
P06734	Mouse	5	Piece	20	5	1050	1000
P07865	1.22 Floppies	5	Piece	100	20	525	500
P07868	Keyboards	2	Piece	10	3	3150	3050
P07885	CD Drive	2.5	Piece	10	3	5250	5100
P07965	540 HDD	4	Piece	10	3	8400	8000
P07975	1.44 Drive	5	Piece	10	3	1050	1000
P08865	1.22 Drive	5	Piece	2	3	1050	1000

Table Name: SALES_MASTER

Salesman_No	Salesman_Name	Address1	Address2	City	Pincode	State	Saleamt	Tgt_To_Get	Ytd sales	Remarks
S00001	Kiran	A/14	Worli	Bombay	400002	MAH	3000	100	50	Good
S00002	Manish	65	Nariman	Bombay	400001	MAH	3000	200	100	Good
S00003	Ravi	P-7	Bandra	Bombay	400032	MAH	3000	200	100	Good
S00004	Ashish	A/5	Juhu	Bombay	400044	MAH	3500	200	150	Good

Table Name: SALES_ORDER

S_ORDER_NO	S_ORDER_DATE	CLIENT_NO	DELY_TYPE	BILLED	SALESMANNO	DELY_DATE	ORDER_STATUS
O19001	12-JAN-96	C00001	F	N	S00001	20-JAN-96	In Process
O19002	25-JAN-96	C00002	P	N	S00002	27-JAN-96	Canceled
O46865	18-FEB-96	C00003	F	Y	S00003	20-FEB-96	Fulfilled
O19003	03-APR-96	C00001	F	Y	S00001	07-APR-96	Fulfilled
O46866	20-MAY-96	C00004	P	N	S00002	22-MAY-96	Canceled
O10008	24-MAY-96	C00005	F	N	S00004	26-MAY-96	In Process

Table Name: SALES_ORDER_DETAIL

S_ORDER_NO	PRODUCT_NO	QTY_ORDERED	QTY_DISP	PRODUCT_RATE
O19001	P00001	4	4	525
O19001	P07965	2	1	8400
O19001	P07885	2	1	5250
O19002	P00001	10	0	525
O46865	P07868	3	3	3150

O46865	P07885	3	1	5250
O46865	P00001	10	10	525
O46865	P03453	4	4	1050
O19003	P03453	2	2	1050
O19003	P06734	1	1	12000
O46866	P07965	1	0	8400
O46866	P07975	1	0	1050
O10008	P00001	10	5	525
O10008	P07975	5	3	1050

Table Name: challan_header

CHALLAN _NO	S_ORDER_NO	CHALLAN_DATE	BILLED
CH9001	O19001	12-DEC-95	Y
CH6865	O46865	12-NOV-95	Y
CH3965	O10008	12-OCT-95	Y

Table Name: challan_details

CHALLAN _NO	PRODUCT _NO	QTY_DISP
CH9001	P00001	4
CH9001	P07965	1
CH9001	P07885	1
CH6865	P07868	3
CH6865	P03453	4
CH6865	P00001	10
CH3965	P00001	5
CH3965	P07975	2

QUERIES

Single Table Retrival :

1.	Find out the names of all the clients.
2.	Print the entire client_master table.
3.	Retrieve the list of names and the cities of all the clients.
4.	List the various products available from the product_master table.
5.	Find the names of all clients having ‘a’ as the second letter in their names.
6.	Find out the clients who stay in a city whose second letter is ‘a’.

7.	Find the list of all clients who stay in city ‘Bombay’ or city ‘Delhi’ or city ‘Madras’.
8.	List all the clients who are located in Bombay.
9.	Print the list of clients whose bal_due are greater than value 10000.
10.	Print the information from sales_order table of orders placed in the month of January
11.	Display the order information for client_no ‘C00001’ and ‘C00002’.
12.	Find the products with description as ‘1.44 Drive and ‘1.22 Drive.’
13.	Find the products whose selling price is greater than 2000 and less than or equal to 5000.
14.	Find the products whose selling price is more than 1500 and also find the new selling price as original selling price*15.
15.	Rename the new column in the above query as new_price.
16.	Find the products whose cost_price is less than 1500.
17.	List the products in sorted order of their description.
18.	Calculate the square root of the price of each product.
19.	Divide the cost of product ‘540 HDD’ by difference between its price and 100.
20.	List the names, city and state of clients not in the state of ‘Maharashtra’.
21.	List the product_no, description, sell_price of products whose description begin with letter ‘M’.
22.	List all the orders that were cancelled in the month of March.
23.	Display all the Client names in Capitals and State in Lowercase.
24.	Price must be displayed in 8,2 format. Put '*' before the amount. e.g. ***23.90.
25.	Display Description and Qty_on_hand for each product.
26.	Display sales transactions (qty_ordered and qty_disp) from Sales_order_details.

Set Functions and Concatenation:

27.	Count the total number of orders.
28.	Calculate the average price of all the products.
29.	Calculate the minimum price of products.

30.	Determine the maximum and minimum product prices. Rename the title as max_price and min_price respectively.
31.	Count the number of products having price greater than or equal to 1500.
32.	Find all the products whose qty_on_hand is less than reorder level.
33.	Print the information of client_master,product_master,sales_order table in the following format for all the records :- {cust_name} has placed order {order_no} on {s_order_date}

Having and Group By:

34.	Print the description and total qty sold for each product.
35.	Find the value of each product sold.
36.	Calculate the average qty sold for each client that has a maximum order value of 15000.00
37.	Find out the total sales amount receivable for the month of jan.It will be the sum total of all the billed orders for the month.
38.	Print the information of product_master,order_detail table in the following format for all the records :- {description} worth Rs. {total sales for the product} was sold.
39.	Print the information of product_master,order_detail table in the following format for all the records :- {description} worth Rs. {total sales for the product} was ordered in the month of {s_order_date in month format}
40.	Find out the total Qty_Disp and Qty_ordered of each item from Sales_order_Details.
41.	Find out the no. of sales transactions of each day from Sales_order_details.
42.	Compute average, minimum and maximum Qty_ordered for each product.

Smt. Chandaben Mohanbhai Patel Institute of Computer Applications
BCA – Semester IV
CA221 – Data Structures and Algorithms

Practical Set – I Programs on Array and Linked List	
1	Write a program to accept n elements and print it.
2	Write a program to accept n elements and find the summation and average of it.
3	Write a program to accept n elements and print all odd elements and count odd elements.
4	Write a program to accept n elements and find the maximum and minimum element from it.
5	Write a program to perform the following operations on a stack. (Implement the stack using array and linked list both) • PUSH • POP • ISEMPTY • ISFULL • PEEP
6	Write a program to perform the following operation on a simple queue. (Implement the queue using array) • Insert an element • Remove an element.
7	Write a program to perform the following operation on a circular queue. (implement the queue using array) • Insert an element • Remove an element
8	Write a program to create a Singly Linked List in LIFO fashion.
9	Write a program to create a Singly Linked List in FIFO fashion.
10	Write a program to create a sorted Singly Linked List.
11	Write program perform the following operations on a Singly Linked List. 1. Insert an element 2. Delete an element 3. Find the sum of elements of the List

	4. Count number of the nodes in the Linked List 5. Search a given element in the Linked List. 6. Reverse the Linked List. 7. Make a copy of the given Linked List 8. Concatenate two Linked List 9. Merge two Linked List. 10. Find the union of the two given Linked List 11. Find the intersection of the two given Linked List.
12	Write a program to create a sorted Doubly Linked List.
13	Write a program to create a Doubly Linked List in LIFO fashion.
14	Write a program to create a Doubly Linked List in FIFO fashion.
15	Write a program perform the following operations on a doubly Linked List. 1. Insert an element 2. Delete an element 3. Find the sum of element of the List 4. Count number of the nodes in the Linked List 5. Search a given element in the Linked List. 6. Reverse the Linked List 7. Make a copy of the given Linked List 8. Concatenate two Linked listed. 9. Merge two Linked List. 10. Find the union of the two given Linked List. 11. Find the intersection of the two given Linked List.
16	Write a program to create a Circular Single Linked List.
17	Write a program to create a Double Linked List.

Practical Set – II Programs on Binary Tree	
1	Write a program to create a binary search tree.
2	Write a program to create a binary search tree and print it's elements in inorder. (write iterative code)
3	Write a program to create a binary search tree and print its elements in preorder. (write iterative code)
4	Write a program to create a binary search tree and print its elements in postorder. (write iterative code)
5	Write a program to make another copy of a given binary search tree.
6	Write a program to count no of leaf nodes in a given binary tree.
7	Write a program to search an element in a given binary search tree.
8	Write a program to accept n elements and check particular element is present how many times. 1. Linear Search 2. Binary Search
9	Write a program to accept n elements and arrange all the elements in Ascending order. 1. Bubble Sort 2. Selection Sort 3. Insertion Sort 4. Shell Sort 5. Quick Sort 6. Merge Sort 7. Radix Sort 8. Heap Sort 9. Topological Sort

ASSIGNMENT-1

Objectives

- **Hands on with Basics of Data Definition Language (DDL) commands Like Create Table, Alter Table, Truncate Table, Drop table etc.**
- **Hands on with Basics of Data Manipulation Language (DML).**
- **To understand and apply constraints to the tables.**

Create Following tables

1. Table1 : stud_master

Column Name	Datatype and Size	Constraints	Description
Sid	varchar(10)	Primary Key. Should contain BCA or MCA	Student ID. Example:21BCA001
Fname	Varchar2(50)		Name of a student. Example: Amit
Lname	Varchar2(50)		Surname of a student. Example: Patel
Branch	varchar2(10)		
Semester	Char(1)		
City	Varchar2(30)		
DOB	DATE		Format: DD-MMM-YYYY. For example: 18-JAN-2003
Gender	CHAR(1)	Should Contain M or m or F or f	

Insert following data in above table:

Sid	Fname	Lname	Branch	Semester	City	DOB	Gender
21BCA001	Vishal	Patel	BCA	2	Anand	18-APR-2002	M
21BCA002	Mahesh	Mandavia	BCA	2		20-JUL-2001	M
20MCA003	Monesh	Viradia	MCA	2	Surat	21-MAR-1998	M
19BCA024	Frida	Pinto	BCA	4	Vasad	18-APR-2002	F
20BCA027	Tanvi	Sharma	BCA	2	Nadiad	04-JAN-2002	F
18BCA045	Vishal	Shah	BCA	4	Anand	08-MAR-2002	M
17MCA060	Shailesh	Macwan	MCA	4		15-APR-2001	M
19BCA100	Rupesh	Misra	BCA	4	Surat	25-AUG-2002	M
17MCA002	Vishal	Mehta	MCA	4	Nadiad	18-OCT-2002	M
17BCA124	Priya	Dave	BCA	4	Vasad	06-FEB-2002	F

2. Table 2: sub_master

Column Name	Datatype and Size	Constraints	Description
Subcode	varchar(10)	Primary Key. Should start with CA	Example:CA119
Subject	Varchar2(40)		subject name
Th_credit	Number(1)	Should be > 0 and <7	Credits in theory portion of the subject

Pr_credit	Number(1)	Should be > 0 and <4	Credits in practical portion of the subject

Insert following data in above table:

Subcode	Subject	Th_credit	Pr_credit
CA119	Database Fundamentals	4	3
CA118	Advanced Programming	4	3
CA117	Operating system concepts	4	3
CA221	Data Structures and Algorithms	4	3
CA321	Basics of Mobile Applications		3
CA115	Management Information System	3	0
CA836	Framework and Applications	0	3
CA835	Artificial Intelligence	3	0

3. Table3 : stud_subject

Column Name	Datatype and Size	Constraints	Description
Sid	varchar(10)	Should contain BCA or MCA	Student ID. Example:21BCA001
Subcode	varchar(10)	Should start with CA	Example:CA119

Insert following data in above table:

Sid	Subcode
21BCA001	CA119
21BCA002	CA119
20MCA003	CA119
19BCA024	CA321
20BCA027	CA221
18BCA045	CA836
17MCA060	CA836
19BCA100	CA835
17MCA002	CA835
17BCA124	CA119
21BCA001	CA119
21BCA002	CA119
20MCA003	CA321

Solve Following queries:

1. Display all student details.
2. Display all subject details.
3. List all the students from Anand city.
4. List all the female students.
5. List all the male students.
6. Display name of the students born in April as per the following format:

Name	DOB
Vishal Patel	18-Apr-2002
Shailesh Macwan	15-Apr-2001

7. List students of BCA branch.
8. List students of MCA branch.
9. List students who have taken admission in year 2017.
10. List BCA students who have taken admission in year 2018.
11. Display details of student with no city.
12. Display gender wise total number of students (Total Male and Total Female).
13. Display details of all students of semester 2.
14. Display age of all the students in following format:

Name	**AGE**
Vishal Patel	16
Shailesh Macwan	17

15. Display all female students of BCA.
16. Display all male students of MCA.
17. Display all distinct branches
18. Display total credits in all subject as per following format:

Subcode	**Subject**	**Total Credits**
CA119	Database Fundamentals	7
CA118	Advanced Programming	7
----	------	-------

19. Display details of Artificial Intelligence subject.
20. Display only theory subjects.
21. Display only practical subjects.
22. Display all studid and subcode combinations.
23. Display studids who are enrolled for subject code CA836.
24. Display all subject codes of studid 17BCA060
25. Display total number of distinct subject codes.
26. Display total number of distinct studids.
27. Display list of students who have not done course registration.
28. Change the city of roll no 21BCA001 to Mumbai.
29. Insert the city of roll no 21BCA002
30. Increase the size of branch column to varchar2(20).
31. Delete the subject 'Framework and Applications'.

Queries of Expressions/Functions

1. Find eldest student. HINT: use max function.
2. Find youngest student. HINT: use min function.
3. Find total number of students. HINT: use count function.
4. Find total credits in all theory subjects. HINT: use sum function.
5. Find total credits in all practical subjects. HINT: use sum function.
6. Display student's DOB in following format: HINT: use to_char() function.

Name	DOB
Vishal Patel	April 18, 2002
Shailesh Macwan	April 15, 2001

7. Display student's DOB in following format: HINT: use add_months() function.

Name	DOB
Vishal Patel	18-Oct-2002
Shailesh Macwan	15-Oct-2001

8. Find difference in months between your DOB and today's date.
9. Add following column in the **stud_master created.**

Mobile	Number(10)		

10. Create following temporary table. Name: **Stud_temp**

Column Name	**Datatype and Size**	**Constraints**	**Description**
Temp_ID	varchar(10)	Not Null	Student ID. Example:21BCA001
Remarks	varchar(20)		

   1. Enter the Data in **Stud_temp** table.
   2. Display content of the **Stud_temp** Table.
   3. Truncate the **Stud_temp table** without explicitly deleting the rows.
   4. Again Display content of the **Stud_temp** Table.
   5. Drop The Table Stud_temp.

ASSIGNMENT-2

Objectives

- **Hands on with Basics of Data Definition Language (DDL)**
- **To apply JOINS, GROUPBY and Advance SQL queries.**

(i) Create tables according to the following definition.
CREATE TABLE DEPOSIT (ACTNO VARCHAR2(5) ,CNAME VARCHAR2(18) , BNAME VARCHAR2(18) , AMOUNT NUMBER(8,2) ,ADATE DATE);

CREATE TABLE BRANCH(BNAME VARCHAR2(18),CITY VARCHAR2(18));

CREATE TABLE CUSTOMERS(CNAME VARCHAR2(19) ,CITY VARCHAR2(18));

CREATE TABLE BORROW(LOANNO VARCHAR2(5), CNAME VARCHAR2(18), BNAME VARCHAR2(18), AMOUNT NUMBER (8,2));

(ii) Insert the data as shown below.

DEPOSIT

ACTNO	CNAME	BNAME	AMOUNT	ADATE
100	ANIL	VRCE	1000.00	1-MAR-95
101	SUNIL	AJNI	5000.00	4-JAN-96
102	MEHUL	KAROLBAGH	3500.00	17-NOV-95
104	MADHURI	CHANDI	1200.00	17-DEC-95
105	PRMOD	M.G.ROAD	3000.00	27-MAR-96
106	SANDIP	ANDHERI	2000.00	31-MAR-96
107	SHIVANI	VIRAR	1000.00	5-SEP-95
108	KRANTI	NEHRU PLACE	5000.00	2-JUL-95
109	MINU	POWAI	7000.00	10-AUG-95

BRANCH

VRCE	NAGPUR
AJNI	NAGPUR
KAROLBAGH	DELHI
CHANDI	DELHI
DHARAMPETH	NAGPUR
M.G.ROAD	BANGLORE
ANDHERI	BOMBAY
VIRAR	BOMBAY
NEHRU PLACE	DELHI
POWAI	BOMBAY

CUSTOMERS

ANIL	CALCUTTA
SUNIL	DELHI
MEHUL	BARODA
MANDAR	PATNA

MADHURI	NAGPUR
PRAMOD	NAGPUR
SANDIP	SURAT
SHIVANI	BOMBAY
KRANTI	BOMBAY
NAREN	BOMBAY

BORROW

LOANNO	CNAME	BNAME	AMOUNT
201	ANIL	VRCE	1000.00
206	MEHUL	AJNI	5000.00
311	SUNIL	DHARAMPETH	3000.00
321	MADHURI	ANDHERI	2000.00
375	PRMOD	VIRAR	8000.00
481	KRANTI	NEHRU PLACE	3000.00

From the above given tables perform the following queries:
(1) Describe deposit, branch.
(2) Describe borrow, customers.
(3) List all data from table DEPOSIT.
(4) List all data from table BORROW.
(5) List all data from table CUSTOMERS.
(6) List all data from table BRANCH.
(7) Give account no and amount of depositors.
(8) Give name of depositors having amount greater than 4000.
(9) Give name of customers who opened account after date '1-12-96'.

2. **Create table and insert sample data in tables**.

Create Table Job (job_id, job_title, min_sal, max_sal)

COLUMN NAME	DATA TYPE
job_id	Varchar2(15)
job_title	Varchar2(30)
min_sal	Number(7,2)
max_sal	Number(7,2)

Create table Employee (emp_no, emp_name, emp_sal, emp_comm, dept_no)

COLUMN NAME	DATA TYPE
emp_no	Number(3)
emp_name	Varchar2(30)
emp_sal	Number(8,2)
emp_comm	Number(6,1)
dept_no	Number(3)

Create table deposit(a_no,cname,bname,amount,a_date).

COLUMN NAME	DATA TYPE

a_no	Varchar2(5)
cname	Varchar2(15)
bname	Varchar2(10)
amount	Number(7,2)
a_date	Date

Create table borrow(loanno,cname,bname,amount).

COLUMN NAME	DATA TYPE
loanno	Varchar2(5)
cname	Varchar2(15)
bname	Varchar2(10)
amount	Varchar2(7,2)

Insert following values in the table **Employee**.

emp_n	emp_name	emp_sal	emp_comm	dept _no
101	Smith	800		20
102	Snehal	1600	300	25
103	Adama	1100	0	20
104	Aman	3000		15
105	Anita	5000	50,000	10
106	Sneha	2450	24,500	10
107	Anamika	2975		30

Insert following values in the table **job**.

job_id	**job_name**	**min_sal**	**max_sal**
IT_PROG	Programmer	4000	10000
MK_MGR	Marketing manager	9000	15000
FI_MGR	Finance manager	8200	12000
FI_ACC	Account	4200	9000
LEC	Lecturer	6000	17000
COMP_OP	Computer Operator	1500	3000

Insert following values in the table **deposit**.

A_no	cname	Bname	Amount	date
101	Anil	andheri	7000	01-jan-06
102	sunil	virar	5000	15-jul-06
103	jay	villeparle	6500	12-mar-06
104	vijay	andheri	8000	17-sep-06

105	keyur	dadar	7500	19-nov-06
106	mayur	borivali	5500	21-dec-06

3. Displaying data from Multiple Tables (join)
(1) Give details of customers ANIL.
(2) Give name of customer who are borrowers and depositors and having living city nagpur
(3) Give city as their city name of customers having same living branch.
(4) Write a query to display the last name, department number, and department name for all employees.
(5) Create a unique listing of all jobs that are in department 30. Include the location of the department in the output
(6) Write a query to display the employee name, department number, and department name for all employees who work in NEW YORK.
(7) Display the employee last name and employee number along with their manager's last name and manager number. Label the columns Employee, Emp#, Manager, and Mgr#, respectively.
(8) Create a query to display the name and hire date of any employee hired after employee SCOTT.

4. To apply the concept of Aggregating Data using Group functions.
(1) List total deposit of customer having account date after 1-jan-96.
(2) List total deposit of customers living in city Nagpur.
(3) List maximum deposit of customers living in bombay.
(4) Display the highest, lowest, sum, and average salary of all employees. Label the columns Maximum, Minimum, Sum, and Average, respectively. Round your results to the nearest whole number.
(5) Write a query that displays the difference between the highest and lowest salaries. Label the column DIFFERENCE.
(6) Create a query that will display the total number of employees and, of that total, the number of employees hired in 1995, 1996, 1997, and 1998
(7) Find the average salaries for each department without displaying the respective department numbers.
(8) Write a query to display the total salary being paid to each job title, within each department.
(9) Find the average salaries > 2000 for each department without displaying the respective department numbers.
(10) Display the job and total salary for each job with a total salary amount exceeding 3000, in which excludes president and sorts the list by the total salary.
(11) List the branches having sum of deposit more than 5000 and located in city bombay.

5. To solve queries using the concept of sub query.
(1) Write a query to display the last name and hire date of any employee in the same department as SCOTT. Exclude SCOTT
(2) Give name of customers who are depositors having same branch city of mr. sunil.
(3) Give deposit details and loan details of customer in same city where pramod is living.
(4) Create a query to display the employee numbers and last names of all employees who earn more than the average salary. Sort the results in ascending order of salary.
(5) Give names of depositors having same living city as mr. anil and having deposit amount greater than 2000
(6) Display the last name and salary of every employee who reports to ford.
(7) Display the department number, name, and job for every employee in the Accounting department.
(8) List the name of branch having highest number of depositors.
(9) Give the name of cities where in which the maximum numbers of branches are located.

(10) Give name of customers living in same city where maximum depositors are located.

6. Manipulating Data

(1) Give 10% interest to all depositors.
(2) Give 10% interest to all depositors having branch vrce
(3) Give 10% interest to all depositors living in nagpur and having branch city bombay.
(4) Write a query which changes the department number of all employees with empno 7788's job to employee 7844'current department number.
(5) Transfer 10 Rs from account of anil to sunil if both are having same branch.
(6) Give 100 Rs more to all depositors if they are maximum depositors in their respective branch.
(7) Delete depositors of branches having number of customers between 1 to 3.
(8) Delete deposit of vijay.
(9) Delete borrower of branches having average loan less than 1000.

ASSIGNMENT-3

Objective:

- **To describe constant, variable, variable initialization, and basic operations on the variables in PL/SQL.**

1. PL/SQL Block

1 Write a PL/SQL block to print 'WELCOME' message.

2 Write a PL/SQL block to input and to print your personal detail in the following format using appropriate variable.

Name: <<Name>>
Address: <<Address>>
City: <<City>>
Mobile: <<Mobile Number>>
Birth Date: <<Birth date>>

3 Write a PL/SQL block to print to initialize two number variables and print the total of it.

4 Write a PL/SQL block to swap the values of two number type of variables. Print the values of variables before and after the swap.

5 Write a PL/SQL block to print minimum of two numbers.

6 WRITE A PL/SQL BLOCK TO FIND MAX OF THREE NUMBERS.

7 WRITE A PL/SQL BLOCK TO PRINT GRADE OF THE STUDENT
INPUT : ROLLNO, MARKS1, MARKS2, MARKS3
CALCULATE TOTAL = MARKS1 + MARKS2 + MARKS3 / 3
BASED ON TOTAL CALCULATE PER
GRADE IS CALCULATED AS PER FORMULA
IF PER >80 GRADE IS AA
IF PER BETWEEN 70 AND 80 GRADE IS A
IF PER BETWEEN 60 TO 70 GRADE IS AB
IF PER BETWEEN 50 TO 60 GRADE IS B
IF PER BETWEEN 40 TO 50 GRADE IS C
IF PER LESS THAN 40 GRADE IS FAIL

8 WRITE A PL/SQL BLOCK TO PRINT 10 NUMBERS USING WHILE AND FOR LOOP

9 WRITE A PL/SQL BLOCK TO PRINT MULTIPLICATION TABLE USING WHILE AND FOR LOOP

10 WRITE A PL/SQL BLOCK TO PRINT FACTORIAL OF GIVEN NUMBER.

ASSIGNMENT-4

Objective:

- **To describe discuss DML statement in PL/SQL with exception handling also.**

PL/SQL Implicit Cursor

1 Write a PL/SQL block to display name and mobile number from EMP table whose id is 101.

2 Write a PL/SQL block to insert 5 numbers in temp table, update the any number by taking input from user, delete any number as per user choice. Also print appropriate message.

3 WRITE A PL/SQL BLOCK TO INSERT ODD NUMBERS IN T1 TABLE AND EVEN NUMBERS IN T2 TABLE
WRITE A PL/SQL BLOCK TO DELETE VALUES FROM TABLE T1
WRITE A PL/SQL BLOCK TO DELETE LAST 50 RECORD OUT OF 100 FROM TEMP TABLE

4 Create table of patient with fields
PATIENT(pid integer, pnm varchar2(10), age integer)
---> TAKE PATIENT ID AS PRIMARY KEY,
AGE SHOULD BE GREATER THAN ZERO
ADD GENDER FIELD INTO PATIENT TABLE AND UPDATE THE RECORD

Write a PL/SQL block which will print
the patient report in the following format
INPUT PATIENT ID FROM USER AND
COMPARE IT WITH TABLE ID AND DISPLAY ACCORDINGLY

patient id: XXX Patient name: XXX
AGE <20 : PRINT IT
AGE >= 20 : PRINT IT
AGE >20 and AGE <=30 : PRINT IT
AGE > 30 & AGE <=40 : PRINT IT
AGE >40 & AGE <60 : PRINT IT
AGE >=60 : DISPLAY IT

5 WRITE A PL/SQL BLOCK TO DO THE FOLLOWING:
INPUT 20 NUMBERS FROM USER
A) FIRST ADD ALL NUMBERS IN T3 TABLE
B) IF NUMBER IS 5,10,15 THEN DO UPDATE T3 (ADD 20 IN)
ELSE DELETE OTHER RECORDS
Print appropriate message for valid and invalid record.

ASSIGNMENT-5

Objective:

- **To describe SELECT-into clause – only one output statement in PL/SQL**

```
CREATE TABLE DOCTOR(DID INTEGER PRIMARY KEY, DNAME
VARCHAR2(10))

CREATE TABLE PATIENT(pid integer PRIMARY KEY,
      patientnm varchar2(10),
      age integer CHECK(AGE > 0),
      GENDER CHAR(1) CHECK(GENDER IN ('M','F','m','f')),
       DID INTEGER REFERENCES DOCTOR)

INSERT 5-7 RECORDS IN BOTH TABLE

SELECT * FROM DOCTOR
SELECT * FROM PATIENT
```

1. Write a PL/SQL block to print senior citizen(maximum age of patient).

2. Write a PL/SQL block to print name of patient who is youngest.

3. WRITE A PL/SQL BLOCK TO COUNT TOTAL NUMBER OF RECORDS FROM PATIENT TABLE

4. WRITE A PL/SQL BLOCK TO DISPLAY MESSAGE OF PATIENT
   E.G. TAKE INPUT AS PID, EXTRACT AGE AND DO THE FOLLOWING

   TAKE AGE FROM TABLE AND BASED ON CONDITION DISPLAY THE RECORD
   IF AGE >= 60 THEN DISPLAY- YOU ARE SENIOR CITIZEN
   IF AGE >40 AND AGE <60 DISPLAY- YOU ARE MATURE CITIZEN
   IF AGE >20 AND AGE <=40 DISPLAY- YOU ARE YOUTH CITIZEN
   IF AGE >14 AND AGE <= 20 DISPLAY - YOU ARE TEENAGER
   IF AGE <=14 DISPLAY YOU ARE CHILD

5 WRITE A PL/SQL BLOCK TO PRINT DEPT NAME OF DEPT ALONG WITH EMPLOYEE NAME WHOSE DEPTNO = 2

ASSIGNMENT-6

Objective:

- **To describe Explicit cursor- ALL records from table or selected records from table and manipulate data based on logic.**

1 Assume that EMP and DEPT table with proper records exists in your database.

EMP(empid, fname, mname, lname, gender, salary, dob, designation, phoneno, deptid)
DEPT(deptid, dname, max_capacity)

Write a PL/SQL block to display all the records.
(use explicit cursor and cursor-For loop)

2 Write a PL/SQL block to display name, phone-no and designation of female employees.

3 Write a PL/SQL block to display highest paid first 5 employees.

4 ADD bonus field in EMP table.

Write a PL/SQL block to calculate bonus based on condition. After calculation of bonus of all the employees, display on screen as well as update it.
IF salary greater than 50k then bonus is 10% of salary
IF salary greater than 25k and less than 50k bonus is 15% of salary
If salary less than 25k then bonus is 20% of salary.

5 Assume that you have EMP and DEPT table with proper records:

Find out the name of employee who has experienced in the Electrical department more than 5 years.

6 Assume that you have EMP and DEPT table with proper records:
Write a PL-SQL block that will print the following report:

```
------------------------------------------------------------------
Id    NAme      Gender   Phone-no    Designation    Deptid   Deptname
------------------------------------------------------------------
1     Mahendra    M    9999222234    Manager       101      MCA
2     Virat       M    9873627789    Sales Manager 102      Civil
3     Rohit       M    8762349087    Clerk          101     MCA
4     Renu        F    7896734567    Accountant     103     Electrical
------------------------------------------------------------------
      Total Records Printed : 4
```

ASSIGNMENT-7

Objective:

- **To discuss Stored Procedure in PL/SQL from block or direct.**

Assume that EMP and DEPT table with proper records exists in your database.

EMP(empid, fname, mname, lname, gender, salary, dob, designation, phoneno, deptid)
DEPT(deptid, dname, max_capacity)

1 Write a Procedure which display average salary of employee.

2 Write a Procedure which pass two parameters, calculate sum of two values and display it in procedure.

3 Write a Procedure which will display total no. of employees whose salary greater than 10000 and execute it.

4 Write a Procedure which will display mobile number of employee whose id is given by user. (pass as parameter)

5 Create a stored procedure that takes and input parameter as department- id and generate an output from the procedure as number of employees who are working in that department.

6 Write a procedure for following task:
When new employee joins in the institute, check if empid is present in the institute or not and the no. of already registered employees *not e*xceeding the max capacity for that institute**.**

7 Write a procedure to display top five highest paid employees whose designation is Officer.

8 Write a procedure to display name of employee from emp table in a given format: (first name, middle name, last name), use CASE statement to have 4 options.
For example,

- Display name in the format " RLP "
- Display name in the format " R.L.Patel "
- Display name in the format " Rambhai L. Patel "
- Display name in the format " Rambhai Laxmanbhai Patel"

9 Enter the employee id from user. Check whether the name exists in table for corresponding id. If yes, Write a PL/SQL procedure to find total no of vowels from employee name. If no, display appropriate message.

10 To vote for Election, write a Procedure which will display the following details:
Eligible to vote(>18 years) || Not Eligible (less than 18 years)
Id & name of employee || id & Name of Employee

ASSIGNMENT-8

Objective:

- **To discuss Stored Function and local function in PL/SQL from block.**

1 Write a function to take two numbers from user and print summation of it.
2 Write a function whether given number is odd or even.
3 Write a function that returns total number of incomplete jobs, using table
JOB(jobid, type_of_job, status)

4 Write a function to which a birthdate is passed and it returns the age in terms of years as of today and a procedure/block which calls this function to display.

5 Write a function which displays the number of items whose weight fall between a given range for a particular color using table
ITEM(itemno, name, color, weight)

6 Create a function to return count of all employees who works in MCA department. This function is called in procedure which displays the output in proper format.

7 ISSUE (ROLLNO, BOOKNO, ISSUE_DATE, RETURN_DATE)

After calculating the fine store the required information in the FINE table so that a report can later be printed out.

FINE (ROLLNO, BOOKNO, ISSUE_DATE, RETURN_DATE, TOT_DAYS, FINE)

Write a procedure or function for the given a table, check which students have to pay fine and the amount of fine to be paid. A student can keep the book for 15 days. If the number of days has exceeded 15 then, the fine is calculated as follows:

Up to 7 days : 50 paise per day

8 – 15 days : Re 1 per day (to be counted from the 8th day onwards)

More than 15 : Re 1.5 per day (to be counted from the 16th day onwards)

8 NEWS_PAPER(news_id, name_of_paper, language, month_frequency, price)

Write a procedure that will display name of paper which is of lowest price.
Write a function which will count total number of newspaper exists language wise.
(English language : 999 papers
Gujarati language : 999 papers)
Hindi language : 999 papers)

ASSIGNMENT-9

Objective:

- **To discuss Database Trigger in PL/SQL.**

Assume above EMP and DEPT table exists with records.

1 Write a code which demonstrate simple trigger.

2 Create a trigger on EMP table that tracks the UPDATE on it. Each record being updated on it shall be tracked and kept as backup in EMP_TRAN table

3 Create a trigger on EMP table that tracks the DELETE on it. Every record being deleted from it shall be kept in EMP_TRAN_BACKUP table.

4 Create a trigger on EMP table that tracks UPDATE being performed on SAL column of the table. If the UPDATE is being performed between 2.30 pm to 3.30 pm then copy the effect on EMP_UPDATE_TRACK table

5 Write a trigger which restrict to enter or update the transaction on Saturday or Sunday
OR
Write a PL\SQL trigger which will not allow to insert records on saturday and sunday.

6 Write a trigger which restrict user to enter name in EMP table either starting with 'X' or is less than 2 letters

7 ADD Blood group in EMP table.

Write a trigger which only accept blood_gr in range(with +ve and –ve blood gr) of A ,B, AB, O.

8 Write a trigger in which accepts empid from user and check whether sal>65000 , don't allow to insert or update emp table.

9 Create a trigger which ensures that Emp name that to be inserted is in capital.

10 Write a trigger which deletes every record from result table if student table is empty.
student(studno,fnm,mnm,lnm,degree,bdth);
result(studno,math,eng ,ss,sci)

11 Write a trigger which accepts student records whose age is between 18 to 30.

12 Consider the table: (Eno, Ename, Basic, Hra, Da, Total)
Assume the following :
* Hra is given in Rupees
* DA is given as percentage as basic
* Total salary will be Basic + da*basic/100 + hra
(i) Write a trigger so that the record on insertion contains basic salary in the range of 5000 to 18000;
DA in the range 16% to 32%; and HRA not more than 3000.

ASSIGNMENT-10

Objective:

- **To discuss PL/SQL Parameterized cursor, Exceptional handling and package.**

1 Item (Item code, item name, unit price)
Bill (bill number, bill date)
Bill_item (bill number, item code, number of units purchased)

Do the following:

(a) For a given bill number, print the bill in the following format.

Bill no.: Bill Date:

Sr. No.	Item code	Item name	Units purchased	unit price	total price
	###	xxxxx	xxxxxxxxxxxxxxxxx	#####	####.##	#######.##

Bill Amount: #########.##

(b) Add one more attribute named bill_amount in the Bill table. Update the Bill table to store bill amount.

2 **PATIENT(pid, pname, age, gender, doctor-id)**
Prepare following report for a Doctor information related to total patient age wise in his hospital.

Gender	Patient in Age Group					Total
	1-12	13-25	25-50	50-100	Above 100	
----	----					----
Total	-------	--------	------	-------	--------	-----

Patient (pid, pame, age, gender , doctor-id)
Appoint(pcode, app_date, time , doctor_id)
Doctor(doctor_id, name, specialization)

Prepare a report from above table taking date, shift, , doctor_id as parameters.
Shift is Morning if time are from 9.00 Am to 1.00 pm and
Shift is Evening if time are from 5.00 to 9.30 pm.

APPOINTMENT SCHEDULE

Date : Doctor name :
Shift : Specialization :

Time	Patient name	Age
9.00		
9.10		
9.20		
…..		

Total no. of patients : ___________

3 ALBUM(album_no, album_name, album_type, release_date, singer_name, no_songs)

Write a procedure which takes parameter as singer name and print the following album details:

Singer name : ______________

Album no	Album name	Type	Release Date	Number of Songs
99	Xx	Xx	Xx	99
99	Xx	Xx	Xx	99

4 Create table and insert proper data.

Tour_Details(tour_no, start_date, group_size, source, destination, vehicle_no, driver_name)

Write a PL/SQL block which will print tour details, a driver is going to take it. (pass driver_no as parameter)

Driver Name : ____________ Vehicle number : __________________

Tour Details

Source	Destination	Group size	Start date	Total days
Xxx	Xxx	999	Xxx	999

5 CUST(cno, accno, name, add, city , phone, email)
ACCOUNT(accno, amount)

Write a stored procedure for banking purpose which perform following Task :

1. Create a new account (insert)
2. Deposit funds (update)
3. Check account balance (view balance)
4. Close the account (delete)

6 Create table and insert proper data.

TRAIN_MAST(train_no, train_name, arrival_time, departure_time, source, destination)

Write a procedure which will print all train details going from Baroda to Bombay
Write a function which will print arrival time and departure time for a given train. (pass train no as a parameter)

7 Given hospital schema

Patient(pat_code, pname, birthdate, gender)

Dentist(dcode, dname, qualification, area, city, experience)

Treatment(dcode, pat_code, treatment_date, treat_details)

Write a procedure or Function to do the following queries: (USE Switch case)

1. Count no. of dentist from ‘lalbaug’ area in baroda city.
2. Display details of dentist with ‘MDS’ as qualification.
3. Display details of treatment given to patient no.’P101’.
4. Display total number of patient who came for treatment as on ’12-Feb-2013’.
5. Display area wise list of dentist who has maximum experience.
6. Display details of youngest dentist.
7. Display dentist details along with their patient.
8. Display details of all female dentist.
9. Count total number of male dentist who live in ‘ahmedabad’.
10. Display patient name without duplicate who come for treatment.
11. Display details of patient who has not given any treatment.

8 **Consider the EMPLOYEE schema as :**

EMPMAST(empno, name, pfno, empbasic , deptno, designation)
HOLIDAY(year, month, holidays, weekoff)
EMPTRAN(empno, month, year, presence, loan(amt))
BANK(accno, empno, bank name, bank branch)

Rules : HRA = 15% of basic
DA = 50% of basic
Medical = 100
PF = 8.33%of basic
Salary is given for (attendance + holidays + weekoff) days.

Print salary slip in foll. Format.

SALARY SLIP of Month :_____ Year _______________

Accno : Name of EMP Designation Dept

Bank name : _________________ Branch : _____________ Dt of Join _________

Presence : __________ Absent days : __________ Sal days : ________

EARNINGS : DEDUCTIONS

BASIC :	LOAN :
DA :	PF :
HRA :	Prof tax : 100
EDICAL : 100	

TOTAL earnings: __________ TOTAL Deductions: __________

NET PAY : ___________

9 Write a code which demonstrate user defined exception.

10 Write a code which demonstrate package in PL/SQL.

================**THERE IS BEGINNING IN THE END !!!**==========================

Practical Assignment – 1 [A]

Revision of C Programming Concepts

1. Write a program to make addition of two numbers and display the summation.
2. Write a program to input Student Name, Branch, Semester, CGPA, and Age and display it.
3. Write a program to display addition, subtraction, multiplication and division of two integers on screen.
4. Write a program to find area of circle. (Area of Circle = PI * r * r)
5. Write a program to check whether the entered number is even or odd. (User Define Function)
6. Write a program to find maximum between two numbers.
7. Write a program to Find Maximum from Three numbers.(User Define Function)
8. Write a program to input 5 numbers in an array and display it.
9. Write a program to input 5 subject marks using array and display total marks and average marks.
10. Write a program to find maximum number from an array.
11. Write a program to input two umbers from user and perform all the basic arithmetic operations. Here use switch case to make menu driven programming.

Practical Assignment – 1 [B]

Getting Familiarity with C++

1. Write a C++ program to print Hello World.
2. Write a C++ program to display any number entered by the user.
3. Write a C++ program to input three numbers and display its total and average.
4. Write a C++ program to enter Day Temperature of 5 cities of Gujarat. Display average temperature.
5. Write a C++ program to find simple interest.
   SI = (P*R*N)/100 with float values
6. Write a C++ program to calculate area of circle.
   Area = π * r * r
7. Write a C++ program to read two numbers from user and print their addition, subtraction, multiplication and division on screen.
8. Write a C++ program to input marks of five subjects of a student and calculate total marks and percentage.
9. Write a C++ program to find maximum between 3 numbers.
10. Write a C++ program to print Student Name, Student Roll Number, Student City, Branch and Mobile Number entered by the user.
11. Write a C++ program to insert 5 numbers in an array and display them.
12. Write a C++ program to insert 5 subject marks of a student in an array, display total marks and percentage on a black screen.
13. Write a C++ program to input temperature of 5 cities in an array and display the temperature in reverse order.
14. Write a C++ program to insert 5 numbers in an array and find maximum number from them.
15. Write a C++ program to insert 5 numbers in an array and find maximum number and minimum number from them.

16. Write a C++ program to input N numbers in an array and display odd numbers and even numbers from an array.
17. Write a C++ program to input N numbers in an array and display sum of odd numbers from an array.
18. Write a C++ program to input N numbers in an array and display sum of even numbers from an array.
19. Write a C++ program to swap two numbers using third variable.

Practical Assignment – 2

Classes and Objects

1. Write a program to input a number in a class Simple and display the number using a member function.
2. Write a program to input a number from user in a class Simple and display the number using a member function.
3. Write a program to input two numbers from user in a class and display summation using a member function.
4. Write a program to input item number, item name and item price from a user into a class item as member function and display the details using a member function.
5. Write a program to create a class ‘Student’ and input Student Name, Branch, Semester, and CGPA from a user into member function and display the details using a member function.
6. WAP to display addition, subtraction, multiplication and division of two integers on screen. Declare Class ‘Calculation’, declare data member num1 and num2 in Private section, write member function for initialize num1 and num2 and write member function for each operation.
7. Write a program to input patient id, patient name, disease name, temperature and weight in a member function defined outside the class, and display the details using member function.
8. Write a program to input 5 subject marks as function arguments entered by a user in main() and display the result and percentage using a member function defined outside the class.
9. Write a program to input 3 numbers as a function argument which has been defined outside the class and find maximum number from them in a member function which is also defined outside the class.
10. Write a program to create a class Employee having data members as name, designation, city, id number, no of days present and monthly salary. Input these details using a member function and display all the details with Yearly salary Income after deducting 10% tax.
11. Write a program to print the area of a rectangle by creating a class named 'Area' having two functions. First function named as 'setDim' takes the length and breadth of the rectangle as parameters and the second function named as 'getArea' returns the area of the rectangle. Length and breadth of the rectangle are entered through keyboard.

Practical Assignment – 3

Array of Objects, Friend Function and Constructors

1. Write a program to input Student Name, Student ID, Branch and Semester for Class Student and display the details of 3 students. (Use array of objects)
2. Write a program to input Item Name, Lot number, Quantity and Price for Class Item and display the details. (Use array of objects)

3. Write a program to take one number from user using a member function and display the number using a friend function.
4. Write a program to input price of two products in a member function and display total bill amount in a friend function.
5. Write a program to input three numbers in a member function and display maximum number using friend function.
6. Write a program to calculate Simple Interest using friend function. The input must be taken in a member function designed outside the class.
7. Write a program to find whether a number entered by user is odd or even using a friend function. The input must be taken using a member function defined outside the class.
8. Write a program to input one number in one class and another number in another class. Now make summation of these two numbers using a friend function.
9. Write a program to input one number in one class and another number in another class. Now largest of these two numbers using a friend function.
10. Write a program to input per day salary and number of days in a class “Salary”. Also input service tax and income tax in a class “Tax”. Design a friend function “Net_Earning” where Total Salary-Total Tax will be calculated.
11. Write a program to initialize two data members with some default values using a constructor and display their summation using member function.
12. Write a program to input height and width values in a default constructor and calculate Area of rectangle using member function.
13. Write a program to make addition of two numbers using a member function. Input of two numbers must be done using a parameterized constructor.
14. Write a program to pass P, R and N in a parametrized constructor and display simple interest using a member function defined outside the class.
15. Write a program to enter number of units consumed for electricity for the months June and July using parameterized constructor having called explicitly. Then, display total bill (no of Units consumed * 7 Rs per Unit) using a member function.
16. Write a program to display addition, subtraction, multiplication and division of two integers on screen. Declare Class “Calculation”. Write Parameterized constructor for initialize num1 and num2. Write member function for each operation.
17. Write a program to initialize one data member with some default constructor and also initialize another data member with parameterized constructor. Now, copy both the data member values by using copy constructor and display them.
18. Write a program to find factorial of any number using a copy constructor.
19. Write a program to calculate addition of two numbers using a copy constructor and make addition in a member function.

Practical Assignment-4

Inheritance and Polymorphism

Lab Assignment || B.Sc. (IT) || SEM II || CA419 : Introduction to Object Oriented Programming

1. Write a program that defines a shape class with a constructor that gives value to width and height. Then define two sub-classes triangle and rectangle, that calculate the area of the shape. In the main, define two objects a triangle and a rectangle and then call the area() function.
2. Write a program with a mother class animal. Inside it define a name and an age variables, and set_value () function. Then create two sub class Zebra and Dolphin which write a message telling the age and name of animal, also giving some extra information for both sub class (e.g. place of origin).place of origin of zebra is Earth and place of origin of dolphin is water.
3. Write a program as per following details

Create one base class Teacher

Data members Name, Department, College name, Email id.

Create sub classes for Math Teacher, English Teacher, and Science Teacher

Data member Qualification, Expertise and salary.

Display following details for each teacher

Name:

Department:

College name:

Email id:

Qualification:

Expertise:

Salary:

4. Write a program as per following details

Create one base class DRUG with following data members

Category- (i.e. stimulants, inhalants, cannabinoids)

Date_of_manufacture, Company name

Create one sub class TABLET derived from DRUG with following data members

Tablet name, Price

Create one sub class PainReliever derived from TABLET with data member

Dosage_units: 1 or 2 or 3

Side_effects : i.e (Nausea, Drowsiness, Dizziness.)

Use_within_days: i.e(10 or 20 or 30).

Use appropriate member function for setting and Getting above details and display details in main function.

5. Write a program as per following details

Create one base class HOTEL with following data members

Hotel_name,

Hotel_type i.e(Three star,five star)

City

Hotel_rate i.e(2000,3000,5000)

Create one base class FLIGHT with following data members

Flight_no

Source city

Destination city

Seat no

Create one sub class PASSENGER derived from HOTEL and FLIGHT with following data members
Name, Age, city
Write appropriate member functions in each class and display all information in main

6. Write a program as per following details
   Create one base class PERSON with following data members
   Name, College name
   Create one sub class STUDENT derived from PERSON with following data members
   Student_id , Marks of five subject, percentage
   Member function:
   showResult( )-Calculate total,percentage and finding class(Dist,First,second,pass)

Create one sub class EMPLOYEE derived from PERSON with following data members
Emp_id, qualification , basic salary
Member function to calculate Net salary and print Net salary
DA=125% of Basic salary
HRA=10% of Basic salary
TA=500
Income tax=5 % of basic salary, if basic salary >50000
Income tax=0, if Basic salary <=50000
Netsalary=(basicsalary+da+hra+ta) - income tax
Write appropriate setter function in each class and display detail of student and employee in main.

Practical Assignment-5

Operator Overloading

1. Write a program to overload unary minus operator
   Ex. Input : obj(10,-20,30) Output : -10,20,-30 (-obj)
2. Write a program to overload pre decrement operator (obj1= --obj)
3. Write a program to overload post decrement operator (obj1=obj--)
4. Write a program to overload Binary *(multiply) operator for object of same class
5. Write a program to overload Binary * (multiply) operator for object of different class
6. Write program to overload = = operator to compare two object of student class.

Practical Assignment – 6

Using Files for IO

1. Write a program to create a file “Product.txt” and write data as Product Name, Product Price and Batch Number in a file.
2. Write a program to append Product Type and Product Weight in a file “Product.txt”.
3. Write a program to create a file “Patient.txt” and enter Patient ID, Patient Name, Disease Name and City in a file. Now read patient details and display all the details on a terminal.
4. Write a program to append Doctor Name and Doctor Speciality in “Patient.txt” file and display the data on a console.

Lab Assignment || B.SC.(IT)|| SEM II || CA420 : FUNDAMENTALS OF OPERATING SYSTEM

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1. Write a command to list the files and directories on the current disk.

2. Write a command to display the version of MS Dos.

3. Write a command to display the current date.

4. Write a command to display the current time.

5. Write a command to create a new file and display the content of that file.

6. Write a command to copy one file to another file.

7. Write a command to rename a file.

8. Write a command to delete a file.

9. Write a command to create directory.

10. Write a command to change root directory to local directory.

11. Write a command to create two directories COMPUTERS and CARS.

12. Write a command to create two directories named as INPUTDEVICE and OUTPUTDEVICE inside COMPUTERS directory.

13. Write a command to create directory structure as follows:

    - COMPUTERS
        - INPUTDEVICES
        - Keyboard.doc
    - OUTPUTDEVICES
        - Monitor.doc
    - CARS
        - Hatchback.doc
        - Suv.txt
        - Sedan

14. Create Keyboard.doc file inside INPUTDEVICES and write something about keyboard in that file.

15. Write a command to display the content of Keyboard.doc file.

16. Write a command that will display directory structure in a tree format.

17. Write a command to delete any text file with confirmation.

18. Write a command to remove OUTPUTDEVICES directory.

19. Write a command to rename Suv.txt

20. Write a command to display attributes of all files

21. Write a command to set read only permission to file

22. Write a command to make any file read only and hidden file

23. Write a command to unhide previous file

24. Write a command to copy one file from one folder to another folder.

25. Write a command to Change the prompt to $ sign

26. Write a command to display information of a system.

1. Create a new directory of following names:

a. BCA

b. BSIT

c. MCA

d. Test

e. Personal

2. Change directory to BCA

3. Create new directory name Lab_1 in BCA

4. Display present working directory

5. Change directory to BCA

6. Create new text file with your id as name (e.g. 18bca001.txt) that stores data like : Your name, You phone no, Your Address

7. Create new text file with name as Cities and store 5 cities name in that.

8. Create a text file with name my_hobbies to store your hobbies.

9. Create a text file with name Numbers to store 1 to 10 numbers.

10. List all files and folders present in BCA folder

11. Transfer all files and folders data to file name file_dir.txt

12. Display data of Cities.txt file.

13. Remove folder with name Test.

14. Copy my_hobbies file to Personal folder.

15. Change name of folder from MCA to MCAL.

16. Add 2 more cities name to file Cities.txt

17. Display data of Cities.txt file.

18. Move file name Cities to MCA folder

19. Display History of used commands.

20. Store History data to file Lab_<No>.txt file.
21. Display current date and time
22. Display date in dd-mm-yy format
23. Display date in mm/dd/yyyy format
24. Display date in dd Mon yyyy format
25. Display day of the week
26. Display time in hh:mm:ss format
27. Display Calander of current year
28. Display Calender of 2020
29. Display Calender of February Month
30. Display Calender of March 2021
31. Transfer current date and time to file cdate.txt
32. Count no. of word of text
33. Count no. of files and folders
34. Count no. of files and folders and save this data to file
35. Count no. of lines and words of Cities.txt file
36. Add 1 more city in file Cities.txt
37. Count no. of lines and words of Cities.txt file
38. Print “Hello World” in Console
39. Print “Hii!! (newline) I am BCA Student “ Message in two lines
40. Create Variable name x and set 5 value in that
41. Create Variable name y and set 10 value in that
42. Print value of variable x on console
43. Print value of variable x and y on console
44. Create a variable “name “and store your name in that.
45. Print your name console using variable
46. Create a variable fname, mname and lname and store appropriate values.
47. Display your full name using variables
48. Create a variable to store 5 subject marks
49. Display 5 subject marks in new line
50. Find total marks and average

51. Display total marks and average on console

52. Create three variables: n1, n2, n3 and store three numbers in that

53. Displays value of n1, n2 and n3

54. Perform addition, multiplication, subtraction and division on variables

1. Write a command to display only last 5 lines of Cities.txt file.

2. Write a command to display only last 3 lines of Numbers.txt file.

3. Write a command to display only last 5 lines of Cities.txt file.

4. Write a command to display only first 3 lines of Numbers.txt file.

5. Write a command to display only first 7 lines of Cities.txt file and transfer that data in separate file.

6. Write a command to find “Ahmedabad” in Cities.txt file.

7. Write a command to Fine all small letters in Hobbles file.

8. Write a command to find all capital letters in your personal data file. (File with data like your ID, Name etc.)

9. Write a command to find “Anand or Ahmedabad” from Cities file

10. Write a command to find your id from personal data file.

11. Write a command to create tow files with following data

a. F1.txt

i. 1 Bob

ii. 2 Jimmy

iii. 3 Tom

b. F2.txt

i. 1 Rocks

ii. 2 Swag

iii. 3 Twin

12. Write a command to join two files : F1.txt and F2.txt

13. Match all lines that contain a vowel

14. Match all lines that contain any of the letters ‘a’, ‘b’, ‘c’, ‘d’ or ‘e’.

15. Match all lines that start with a digit following zero or more spaces. E.g: “ 1.” or “2.

16. Sort with default ordering in cities.txt

17. Sort in reverse ordering in cities.txt

Lab Assignment || B.SC.(IT)|| SEM II || CA420 : FUNDAMENTALS OF OPERATING SYSTEM

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1. WAP to calculate the area & perimeter of a rectangle.

(Hint: Area of Rectangle: l * w; Perimeter: 2 * (l+w) ;)

2. WAP to calculate tax, given the following condition

•If income is less than 1,50,000 then no tax.

•If income is in range of 1,50,001 to 3,00,000 then charge 10% tax.

•If income is in range of 3,00,001 to 5,00,000 then charge 20% tax.

•If income is above 5,00,001 then charge 30% tax.

3. WAP that read a number between 1 to 7 (including both) and then print corresponding day name from the week. (Use switch case)

For example,

•if entered number is 1, then print "Today is Monday",

•if entered number is 2, then print "Today is Tuesday",

•if entered number is 3, then print "Today is Wednesday", and so on.

4. WAP that categorizes a letter as a vowel or display 'it is not a vowel'. (Use switch case.)

5. WAP that read employee basic salary. Calculate gross salary and display appropriate output.

- GS = BS+DA+HRA+MA-PT
- BS=basic Salary
- DA=Dearness Allowance 5% of BS
- HRA=House Rent allowance 10% of BS
- MA=Medical allowance if BS > 50000 then Rs. 500 fix elsr Rs. 300
- PT=Professional Tax Rs. 200 fixs

6. WAP to find out maximum of 3 numbers.

1. An electric power distribution company charges its domestic consumers as follows:

Consumption Units	Rate of Charge
0-100	Rs. 3 per unit
101-300	Rs. 5 per unit
301-500	Rs. 6 per unit
501 and above	Rs. 500 + Rs. 6 per unit

Write a shell script that read customer number & power consumed and prints the amount to be paid by the customer.

2. The commission a life-insurance sales women earns on insurance policies sold is as follows:

Policy Amount (rs)	Commission
Less or equal to 10,000	500 Rs
Between 10,001 to 24,999	700 Rs
Greater than or equal to 25,000	500 Rs + 1% of the amount exceeding 25,000

Write a shell script that reads the amount of insurance sold and outputs the commission due to the sales women.

3. Write a shell script to print following pattern

```
*
**
***
****
*****
```

4. Write a shell script to print following pattern

```
1
22
333
```

4444

55555

5. Write a shell script to print the following pattern:

1

12

123

1234

12345

6. Write a shell script to print following pattern

*

**

***

****

*****

******

*****

****

***

**

*

7. Write a shell script to perform basic arithmetic operations using switch case. (Menu driven)

8. WAP to find sum of first 10 numbers using for loop.

9. WAP to find sum of first 10 numbers using while loop.

10. WAP that displays all numbers between 30 to 20. (Use a for loop and the decrement operator.)

11. WAP to print sum of odd and even digit of first 50 numbers using while loop.

12. WAP to find reverse number.

13. WAP to find number is palindrome or not. (Hint: 111: if we reverse it then also number remains same)

14. WAP to display following pattern.

```
       #
      ###
     #####
    #######
   #########
  ###########
 #############
```

14. Write a Bash program to check if the Number is a Prime or not

1. Write a shell script to calculate profit or loss on sales of product.

a. Product purchase price : 500

b. Product Selling price : 600

c. Profit / Loss : Profit , 100

d. Profit % :20%

2. Write a shell script to enter the marks of a student in four subjects. Then calculate the total, percentage and display the grades obtained by a student.

Percentage >= 90% : Grade A

Percentage >= 80% : Grade B

Percentage >= 70% : Grade C

Percentage >= 60% : Grade D

Percentage >= 40% : Grade E

Percentage < 40% : Grade F

3. Write a shell script to read gender (M/F) and print corresponding gender using switch.

4. Write a shell script to make sum of digits of the number entered by the user. (121 = 4)

5. Write a shell script to print multiplication table.

6. Write a shell script to find factorial of a number.

7. Write a shell script to find Fibonacci series. (for 0 to 100)

a. 0, 1, 1, 2, 3, 5, 8, 13, 21, 34…

ASSIGNMENT-1

Objectives

- **Hands on with Basics of Data Definition Language (DDL) commands**
- **Hands on with Basics of Data Manipulation Language (DML).**
- **To understand and apply constraints to the tables.**
- **To apply JOINS and GROUPBY with aggregate functions.**

a. **TABLE NAME: EMP**

Field Name	Data Type	Null
EMPNO	NUMBER(4)	
ENAME	VARCHAR2(10)	
JOB	VARCHAR2(9)	
MGR	NUMBER(4)	
HIREDATE	DATE	
SAL	NUMBER(7,2)	
COMM	NUMBER(7,2)	
DEPTNO	NUMBER(2)	

TABLE NAME : DEPT

Field Name	Data Type	Null
DEPTNO	NUMBER(2)	
DNAME	VARCHAR2(14)	
LOC	VARCHAR2(13)	

b. **TABLE NAME : SALGRADE**

Field Name	Data Type	Null
GRADE	NUMBER(2)	
LOSAL	NUMBER (4)	
HISAL	NUMBER (4)	

1. Insert following records in respective tables.

Table Name: EMP

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7369	SMITH	CLERK	7902	17-DEC-80	800		20
7499	ALLEN	SALESMAN	7698	20-FEB-81	1600	300	30
7521	WARD	SALESMAN	7698	22-FEB-81	1250	500	30
7566	JONES	MANAGER	7839	02-APR-81	2975		20
7654	MARTIN	SALESMAN	7698	28-SEP-81	1250	1400	30
7698	BLAKE	MANAGER	7839	01-MAY-81	2850		30
7782	CLARK	MANAGER	7839	09-JUN-81	2450		10
7788	SCOTT	ANALYST	7566	09-DEC-81	3000		20
7839	KING	PRESIDENT		17-NOV-81	5000		10
7844	TURNER	SALESMAN	7698	08-SEP-81	1500	0	30
7876	ADAMS	CLERK	7788	12-JAN-83	1100		20
7900	JAMES	CLERK	7698	03-DEC-81	950		30
7802	FORD	ANALYST	7566	03-DEC-81	3000		20
7934	MILLER	CLERK	7782	23-JAN-82	1300		10

Table Name: DEPT

DEPTNO	DNAME	LOC
10	ACCOUNTING	NEW YORK
20	RESEARCH	DALLA
30	SALES	CHICAGO
40	OPERATIONS	BOSTON

Table Name: SALGRADE

GRADE	HISAL	LOSAL
1	700	1200
2	1201	1400
3	1401	2000
4	2001	3000
5	3001	9999

2. Solve the following queries.

1.	Find out the name of all employees.
2.	Display all information of employee.
3.	Display all information of employee whose job type is “SALESMAN”.
4.	Display all information of employee whose Dept No is 20.
5.	Display employee information whose hiredate is in month of DEC.

6.	Display employee information whose salary is greater than 3000.
7.	Find the names of all employees having ‘A’ as the second letter in their names.
8.	Display department information.
9.	Display Salary Grade Information.
10.	Display Name, Job of employee whose manager name is “BLAKE”.
11.	Display Name, Salary of employee whose commission is null.
12.	Display Name, Salary, Hiredate of employee whose Hiredate is in the year of 81.
13.	Update salary by 5% whose department no is 10.
14.	Update commission comm=sal*5% whose job type is “MANAGER”.
15.	Display employee name whose name start with ‘A’.
16.	Display employee details that have maximum salary.
17.	Display employee details that have minimum salary.
18.	Display average salary department wise.
19.	Display employee details whose salary is between 1000 to 2000.
20.	Display employee information department wise.
21.	Display your name in upper, lower and title case.
22.	Display all Employee names and their length after trimming them
23.	Extract first 3 characters from your name and display the same.
24.	Extract last 4 characters from your name and display the same.
25.	Extract only 3 to 7 characters of your full name.
26.	Find out the absolute values of the following: -20, 30, 25, -400, -13, -11, 34, -99.
27.	Update the employee table so each employee name will contain XXXXX at the end of the same.
28.	Now display the same such that they won't display XXXXX at the end of each name.
29.	Round the following numbers by 5, 2, 1, 3, 4 decimal places: 4.091978,232.837733, 822.2325443.

ASSIGNMENT-2

Objective:

- **To describe constant, variable, variable initialization, and basic operations on the variables in PL/SQL.**

1. PL/SQL Block

1 Write a PL/SQL block to print ‘WELCOME’ message.

2 Write a PL/SQL block to input and to print your personal detail in the following format using appropriate variable.

Name: <<Name>>
Address: <<Address>>
City: <<City>>
Mobile: <<Mobile Number>>
Birth Date: <<Birth date>>

3 Write a PL/SQL block to print to initialize two number variables and print the total of it.

4 Write a PL/SQL block to swap the values of two number type of variables. Print the values of variables before and after the swap.

5 Write a PL/SQL block to print minimum of two numbers.

6 WRITE A PL/SQL BLOCK TO FIND MAX OF THREE NUMBERS.

7 WRITE A PL/SQL BLOCK TO PRINT GRADE OF THE STUDENT
INPUT : ROLLNO, MARKS1, MARKS2, MARKS3
CALCULATE TOTAL = MARKS1 + MARKS2 + MARKS3 / 3
BASED ON TOTAL CALCULATE PER
GRADE IS CALCULATED AS PER FORMULA
IF PER >80 GRADE IS AA
IF PER BETWEEN 70 AND 80 GRADE IS A
IF PER BETWEEN 60 TO 70 GRADE IS AB
IF PER BETWEEN 50 TO 60 GRADE IS B
IF PER BETWEEN 40 TO 50 GRADE IS C
IF PER LESS THAN 40 GRADE IS FAIL

8 WRITE A PL/SQL BLOCK TO PRINT 10 NUMBERS USING WHILE AND FOR LOOP

9 WRITE A PL/SQL BLOCK TO PRINT MULTIPLICATION TABLE
USING WHILE AND FOR LOOP

10 WRITE A PL/SQL BLOCK TO PRINT FACTORIAL OF GIVEN NUMBER.

ASSIGNMENT-3

Objective:

- **To describe discuss DML statement in PL/SQL with exception handling also.**

PL/SQL Implicit Cursor

1 Write a PL/SQL block to display name and mobile number from EMP table whose id is 101.

2 Write a PL/SQL block to insert 5 numbers in temp table, update the any number by taking input from user, delete any number as per user choice. Also print appropriate message.

3 WRITE A PL/SQL BLOCK TO INSERT ODD NUMBERS IN T1 TABLE AND EVEN NUMBERS IN T2 TABLE
WRITE A PL/SQL BLOCK TO DELETE VALUES FROM TABLE T1
WRITE A PL/SQL BLOCK TO DELETE LAST 50 RECORD OUT OF 100 FROM TEMP TABLE

4 Create table of patient with fields
PATIENT(pid integer, pnm varchar2(10), age integer)
---> TAKE PATIENT ID AS PRIMARY KEY,
AGE SHOULD BE GREATER THAN ZERO
ADD GENDER FIELD INTO PATIENT TABLE AND UPDATE THE RECORD

Write a PL/SQL block which will print
the patient report in the following format
INPUT PATIENT ID FROM USER AND
COMPARE IT WITH TABLE ID AND DISPLAY ACCORDINGLY

patient id: XXX Patient name: XXX
AGE <20 : PRINT IT
AGE >= 20 : PRINT IT
AGE >20 and AGE <=30 : PRINT IT
AGE > 30 & AGE <=40 : PRINT IT
AGE >40 & AGE <60 : PRINT IT
AGE >=60 : DISPLAY IT

5 WRITE A PL/SQL BLOCK TO DO THE FOLLOWING:
INPUT 20 NUMBERS FROM USER
A) FIRST ADD ALL NUMBERS IN T3 TABLE
B) IF NUMBER IS 5,10,15 THEN DO UPDATE T3 (ADD 20 IN)
ELSE DELETE OTHER RECORDS
Print appropriate message for valid and invalid record.

ASSIGNMENT-4

Objective:

- **To describe SELECT-into clause – only one output statement in PL/SQL**

CREATE TABLE DOCTOR(DID INTEGER PRIMARY KEY, DNAME VARCHAR2(10))

CREATE TABLE PATIENT(pid integer PRIMARY KEY,
patientnm varchar2(10),
age integer CHECK(AGE > 0),
GENDER CHAR(1) CHECK(GENDER IN ('M','F','m','f')),
DID INTEGER REFERENCES DOCTOR)

INSERT 5-7 RECORDS IN BOTH TABLE

SELECT * FROM DOCTOR
SELECT * FROM PATIENT

1. Write a PL/SQL block to print senior citizen(maximum age of patient).

2. Write a PL/SQL block to print name of patient who is youngest.

3. WRITE A PL/SQL BLOCK TO COUNT TOTAL NUMBER OF RECORDS FROM PATIENT TABLE

4. WRITE A PL/SQL BLOCK TO DISPLAY MESSAGE OF PATIENT E.G. TAKE INPUT AS PID, EXTRACT AGE AND DO THE FOLLOWING

   TAKE AGE FROM TABLE AND BASED ON CONDITION DISPLAY THE RECORD
   IF AGE >= 60 THEN DISPLAY- YOU ARE SENIOR CITIZEN
   IF AGE >40 AND AGE <60 DISPLAY- YOU ARE MATURE CITIZEN
   IF AGE >20 AND AGE <=40 DISPLAY- YOU ARE YOUTH CITIZEN
   IF AGE >14 AND AGE <= 20 DISPLAY - YOU ARE TEENAGER
   IF AGE <=14 DISPLAY YOU ARE CHILD

5 WRITE A PL/SQL BLOCK TO PRINT DEPT NAME OF DEPT ALONG WITH EMPLOYEE NAME WHOSE DEPTNO = 2

ASSIGNMENT-5

Objective:

- **To describe Explicit cursor- ALL records from table or selected records from table and manipulate data based on logic.**

1 Assume that EMP and DEPT table with proper records exists in your database.

EMP(empid, fname, mname, lname, gender, salary, dob, designation, phoneno, deptid)
DEPT(deptid, dname, max_capacity)

Write a PL/SQL block to display all the records.
(use explicit cursor and cursor-For loop)

2 Write a PL/SQL block to display name, phone-no and designation of female employees.

3 Write a PL/SQL block to display highest paid first 5 employees.

4 ADD bonus field in EMP table.

Write a PL/SQL block to calculate bonus based on condition. After calculation of bonus of all the employees, display on screen as well as update it.
IF salary greater than 50k then bonus is 10% of salary
IF salary greater than 25k and less than 50k bonus is 15% of salary
If salary less than 25k then bonus is 20% of salary.

5 Assume that you have EMP and DEPT table with proper records:

Find out the name of employee who has experienced in the Electrical department more than 5 years.

6 Assume that you have EMP and DEPT table with proper records:
Write a PL-SQL block that will print the following report:

Id	NAme	Gender	Phone-no	Designation	Deptid	Deptname
1	Mahendra	M	9999222234	Manager	101	MCA
2	Virat	M	9873627789	Sales Manager	102	Civil
3	Rohit	M	8762349087	Clerk	101	MCA
4	Renu	F	7896734567	Accountant	103	Electrical

Total Records Printed : 4

ASSIGNMENT-6

Objective:

- **To discuss Stored Procedure in PL/SQL from block or direct.**

Assume that EMP and DEPT table with proper records exists in your database.

EMP(empid, fname, mname, lname, gender, salary, dob, designation, phoneno, deptid)
DEPT(deptid, dname, max_capacity)

1 Write a Procedure which display average salary of employee.

2 Write a Procedure which pass two parameters, calculate sum of two values and display it in procedure.

3 Write a Procedure which will display total no. of employees whose salary greater than 10000 and execute it.

4 Write a Procedure which will display mobile number of employee whose id is given by user. (pass as parameter)

5 Create a stored procedure that takes and input parameter as department- id and generate an output from the procedure as number of employees who are working in that department.

6 Write a procedure for following task:
When new employee joins in the institute, check if empid is present in the institute or not and the no. of already registered employees *not e*xceeding the max capacity for that institute**.**

7 Write a procedure to display top five highest paid employees whose designation is Officer.

8 Write a procedure to display name of employee from emp table in a given format:
(first name, middle name, last name), use CASE statement to have 4 options.
For example,

- Display name in the format “ RLP “
- Display name in the format “ R.L.Patel “
- Display name in the format “ Rambhai L. Patel “
- Display name in the format “ Rambhai Laxmanbhai Patel”

9 Enter the employee id from user. Check whether the name exists in table for corresponding id. If yes, Write a PL/SQL procedure to find total no of vowels from employee name. If no, display appropriate message.

10 To vote for Election, write a Procedure which will display the following details:
Eligible to vote(>18 years) || Not Eligible (less than 18 years)
Id & name of employee || id & Name of Employee

ASSIGNMENT-7

Objective:

- **To discuss Stored Function and local function in PL/SQL from block.**

1 Write a function to take two numbers from user and print summation of it.
2 Write a function whether given number is odd or even.
3 Write a function that returns total number of incomplete jobs, using table
JOB(jobid, type_of_job, status)

4 Write a function to which a birthdate is passed and it returns the age in terms of years as of today and a procedure/block which calls this function to display.

5 Write a function which displays the number of items whose weight fall between a given range for a particular color using table
ITEM(itemno, name, color, weight)

6 Create a function to return count of all employees who works in MCA department. This function is called in procedure which displays the output in proper format.

7 ISSUE (ROLLNO, BOOKNO, ISSUE_DATE, RETURN_DATE)

After calculating the fine store the required information in the FINE table so that a report can later be printed out.

FINE (ROLLNO, BOOKNO, ISSUE_DATE, RETURN_DATE, TOT_DAYS, FINE)

Write a procedure or function for the given a table, check which students have to pay fine and the amount of fine to be paid. A student can keep the book for 15 days. If the number of days has exceeded 15 then, the fine is calculated as follows:

Up to 7 days : 50 paise per day

8 – 15 days : Re 1 per day (to be counted from the 8th day onwards)

More than 15 : Re 1.5 per day (to be counted from the 16th day onwards)

8 NEWS_PAPER(news_id, name_of_paper, language, month_frequency, price)

Write a procedure that will display name of paper which is of lowest price.
Write a function which will count total number of newspaper exists language wise.
(English language : 999 papers
Gujarati language : 999 papers)
Hindi language : 999 papers)

ASSIGNMENT-8

Objective:

- **To discuss Database Trigger in PL/SQL.**

Assume above EMP and DEPT table exists with records.

1 Write a code which demonstrate simple trigger.

2 Create a trigger on EMP table that tracks the UPDATE on it. Each record being updated on it shall be tracked and kept as backup in EMP_TRAN table

3 Create a trigger on EMP table that tracks the DELETE on it. Every record being deleted from it shall be kept in EMP_TRAN_BACKUP table.

4 Create a trigger on EMP table that tracks UPDATE being performed on SAL column of the table. If the UPDATE is being performed between 2.30 pm to 3.30 pm then copy the effect on EMP_UPDATE_TRACK table

5 Write a trigger which restrict to enter or update the transaction on Saturday or Sunday
OR
Write a PL\SQL trigger which will not allow to insert records on saturday and sunday.

6 Write a trigger which restrict user to enter name in EMP table either starting with 'X' or is less than 2 letters

7 ADD Blood group in EMP table.

Write a trigger which only accept blood_gr in range(with +ve and –ve blood gr) of A ,B, AB, O.

8 Write a trigger in which accepts empid from user and check whether sal>65000 , don't allow to insert or update emp table.

9 Create a trigger which ensures that Emp name that to be inserted is in capital.

10 Write a trigger which deletes every record from result table if student table is empty.
student(studno,fnm,mnm,lnm,degree,bdth);
result(studno,math,eng ,ss,sci)

11 Write a trigger which accepts student records whose age is between 18 to 30.

12 Consider the table: (Eno, Ename, Basic, Hra, Da, Total)
Assume the following :
* Hra is given in Rupees
* DA is given as percentage as basic
* Total salary will be Basic + da*basic/100 + hra
(i) Write a trigger so that the record on insertion contains basic salary in the range of 5000 to 18000;
DA in the range 16% to 32%; and HRA not more than 3000.

ASSIGNMENT-9

Objective:

- **To discuss PL/SQL Parameterized cursor, Exceptional handling and package.**

1 Item (Item code, item name, unit price)
Bill (bill number, bill date)
Bill_item (bill number, item code, number of units purchased)

Do the following:

(a) For a given bill number, print the bill in the following format.

Bill no.: Bill Date:

Sr. No.	Item code	Item name	Units purchased	unit price	total price
###	xxxxx	xxxxxxxxxxxxxxxxx	#####	####.##	#######.##
				Bill Amount:	#########.##

(b) Add one more attribute named bill_amount in the Bill table. Update the Bill table to store bill amount.

2 **PATIENT(pid, pname, age, gender, doctor-id)**
Prepare following report for a Doctor information related to total patient age wise in his hospital.

Gender	Patient in Age Group 1-12	13-25	25-50	50-100	Above 100	Total
----	----					----
Total	-------	--------	------	-------	--------	-----

Patient (pid, pame, age, gender , doctor-id)
Appoint(pcode, app_date, time , doctor_id)
Doctor(doctor_id, name, specialization)
Prepare a report from above table taking date, shift, , doctor_id as parameters.
Shift is Morning if time are from 9.00 Am to 1.00 pm and
Shift is Evening if time are from 5.00 to 9.30 pm.

APPOINTMENT SCHEDULE

Date : Doctor name :
Shift : Specialization :

Time	Patient name	Age
9.00		
9.10		
9.20		
…..		

Total no. of patients : ___________

3 ALBUM(album_no, album_name, album_type, release_date, singer_name, no_songs)

Write a procedure which takes parameter as singer name and print the following album details:

Singer name : ______________

Album no	Album name	Type	Release Date	Number of Songs
99	Xx	Xx	Xx	99
99	Xx	Xx	Xx	99

4 Create table and insert proper data.

Tour_Details(tour_no, start_date, group_size, source, destination, vehicle_no, driver_name)

Write a PL/SQL block which will print tour details, a driver is going to take it. (pass driver_no as parameter)

Driver Name : ____________ Vehicle number : __________________

Tour Details

Source	Destination	Group size	Start date	Total days
Xxx	Xxx	999	Xxx	999

5 CUST(cno, accno, name, add, city , phone, email)
ACCOUNT(accno, amount)
Write a stored procedure for banking purpose which perform following Task :

1. Create a new account (insert)
2. Deposit funds (update)
3. Check account balance (view balance)
4. Close the account (delete)

6 Create table and insert proper data.

TRAIN_MAST(train_no, train_name, arrival_time, departure_time, source, destination)

Write a procedure which will print all train details going from Baroda to Bombay
Write a function which will print arrival time and departure time for a given train. (pass train no as a parameter)

ASSIGNMENT-10

1 Given hospital schema

Patient(pat_code, pname, birthdate, gender)

Dentist(dcode, dname, qualification, area, city, experience)

Treatment(dcode, pat_code, treatment_date, treat_details)

Write a procedure or Function to do the following queries: (USE Switch case)

1. Count no. of dentist from ‘lalbaug’ area in baroda city.
2. Display details of dentist with ‘MDS’ as qualification.
3. Display details of treatment given to patient no.’P101’.
4. Display total number of patient who came for treatment as on ’12-Feb-2013’.
5. Display area wise list of dentist who has maximum experience.
6. Display details of youngest dentist.
7. Display dentist details along with their patient.
8. Display details of all female dentist.
9. Count total number of male dentist who live in ‘ahmedabad’.
10. Display patient name without duplicate who come for treatment.
11. Display details of patient who has not given any treatment.

2 **Consider the EMPLOYEE schema as :**

EMPMAST(empno, name, pfno, empbasic , deptno, designation)
HOLIDAY(year, month, holidays, weekoff)
EMPTRAN(empno, month, year, presence, loan(amt))
BANK(accno, empno, bank name, bank branch)

Rules : HRA = 15% of basic
DA = 50% of basic
Medical = 100
PF = 8.33%of basic
Salary is given for (attendance + holidays + weekoff) days.

Print salary slip in foll. Format.

SALARY SLIP of Month :_____ Year _______________

Accno : Name of EMP Designation Dept

Bank name : _________________ Branch : _____________ Dt of Join _________

Presence : __________ Absent days : __________ Sal days : ________

EARNINGS : DEDUCTIONS

BASIC :	LOAN :
DA :	PF :
HRA :	Prof tax : 100
EDICAL : 100	

TOTAL earnings: ___________ TOTAL Deductions: ___________

NET PAY : ____________

---

3 Write a code which demonstrate user defined exception.

4 Write a code which demonstrate package in PL/SQL.

================**THE END, NO the Journey doesn't end here !!!**===================

CA615: Embedded Programming using Raspberry PI

List of Practical Assignment

Sr No	Title of the Experiment
1.	Implementing digital components and circuits using Tinker cad.
2.	Connecting and designing various circuits using Tinker cad.
3.	Introduction and installation of Raspberry PI Board and its OS.
4.	Connecting Raspberry Pi virtually using VNC Viewer, SSH and Remote desktop application.
5.	Introduction to Python for Raspberry PI.
6.	Python Programming for Raspberry PI: Basics.
7.	Python Programming for Raspberry PI: Operators, Decision making and loops.
8.	Python Programming for Raspberry PI: Numbers and strings.
9.	Python Programming for Raspberry PI: List, Tuple and dictionary.
10.	Python Programming for Raspberry PI: Functions.
11.	Interfacing Raspberry PI with sensors.
12.	Interfacing Raspberry PI with camera modules.

EPR Python Lab programs:

1. Python program to add two numbers.
2. Python program to add two numbers provided by the user
3. Python program to add two numbers using function
4. Python program to multiply three number
5. Python program to multiply three numbers using function
6. Python program to find average of five numbers
7. Python program to find square root of the number
8. python program to find area of circle
9. python program to find the area of the right-angle triangle
10. Python program to calculate simple interest using function
11. Python Program to get the first digit of number
12. Python Program to reverse number
13. Python Program to swap two numbers without using temporary variable
14. Python program to make a simple calculator
15. Python program to print fibonacci series up to n^{th} term
16. Python program to print multiplication table
17. Python program to find factors of a number
18. Python program to find the factorial of a number
19. Python program to convert decimal to binary
20. Python program to convert decimal to hexadecimal
21. Python program to print numbers from 1 to 10 using while loop
22. Python program to print numbers from 1 to 10 using for loop.
23. Python program to create list, edit the list, copy the list, and access items from the list.
24. Python program to create nested list, edit the list, copy the list, and access items from the list.
25. Python program to create tuple and access items from the tuple. Also write code for tuple unpacking.
26. Python program to split an email address into user name and domain using Tuple.
27. Python program to create list slicing and tuple slicing.
28. Python program to create set, add and remove items from the set.
29. Python program to explain various set operations.
30. Python program to create dictionary using various methods.
31. Python program to add and update dictionary and merge two dictionaries.
32. Python program to create nested dictionary.
33. Python program for linear searching.
34. Python program for binary searching.

Python programs:

1.
```
# python program to add two numbers

# take inputs
num1 = 5
num2 = 10

# add two numbers
sum = num1 + num2

# displaying the addition result
print('{0} + {1} = {2}'.format(num1, num2, sum))
```

2.
```
# python program to add two numbers provided by the user

# store input numbers
num1 = input('Enter First Number: ')
num2 = input('Enter Second Number: ')

# add two numbers
# User might also enter float numbers
sum = float(num1) + float(num2)

# displaying the adding result
# value will print in float
print('The sum of numbers {0} and {1} is {2}'.format(num1, num2, sum))
```

3.
```
# Python program to add two numbers using function

def add_num(a,b):   #user-defined function
   sum = a + b   #adding numbers
   return sum   #return value

#taking input from the user
num1 = float(input('Enter first number : '))
num2 = float(input('Enter second number : '))

#function call
print('The sum of numbers {0} and {1} is {2}'.format(
            num1, num2, add_num(num1, num2)))
```

4.

```
# Python program to add two numbers in one line
# Without using any variables

print('The sum is %.2f' %(float(input('Enter First Number: '))
        + float(input('Enter Second Number: '))))
```

5.

```
# Python program to multiply three number

# take inputs
num1 = float(input('Enter first number: '))
num2 = float(input('Enter second number: '))
num3 = float(input('Enter third number: '))

# calculate product
product = num1 * num2 * num3

# print multiplication value
print("The product of number: %0.2f" %product)
```

6.

```
# Python program to multiply three numbers using function

def product_num(num1, num2, num3):  #user-defind function
   num = (num1 * num2 * num3)   #calculate product
   return num   #return value

# take inputs
num1 = float(input('Enter first number: '))
num2 = float(input('Enter second number: '))
num3 = float(input('Enter third number: '))

# function call
product = product_num(num1, num2, num3)

# print multiplication value
print("The product of number: %0.2f" %product)
```

7.

```
# Python program to find average of five numbers

# take inputs
num1 = float(input('Enter first number: '))
```

```
num2 = float(input('Enter second number: '))
num3 = float(input('Enter third number: '))
num4 = float(input('Enter four number: '))
num5 = float(input('Enter fifth number: '))

# calculate average
avg = (num1 + num2 + num3 + num4 + num5) / 5

# print average value
print('The average of numbers = %0.2f' %avg)
```

8.

```
# Python program to find the average of five numbers

# denotes total sum of five numbers
total_sum = 0

for n in range (5):
   # take inputs
   num = float(input('Enter number: '))
   # calculate total sum of numbers
   total_sum += num

# calculate average of numbers
avg = total_sum / 5

# print average value
print('Average of numbers =', avg)
```

9.

```
# Python program to find square root of the number

# take inputs
num = float(input('Enter the number: '))

# calculate square root
sqrt = num ** 0.5

# display result
print('Square root of %0.2f is %0.2f '%(num, sqrt))
```

10.

```
# python program to find area of circle

# store input
r = float(input('Enter the radius of the circle: '))
```

```
# calculate area of circle
area = 3.14 * r * r

# display result
print('Area of circle = %.2f ' %area)
```

11.

```
# python program to find the area of the right-angle triangle

# take inputs
base = float(input('Enter the base of the triangle: '))
height = float(input('Enter the height of the triangle: '))

# calculate area of triangle
area = (1/2) * base * height

# display result
print('Area of triangle = ',area)
```

12.

```
# Python program to calculate simple interest using function

def calculate_simple_interest(P, R, T):
    # calculate simple interest
    SI = (P * R * T) / 100
    return SI;

if __name__ == '__main__':
    # store the inputs
    P = float(input('Enter principal amount: '))
    R = float(input('Enter the interest rate: '))
    T = float(input('Enter time: '))

    # calling function
    simple_interest = calculate_simple_interest(P, R, T)
    # display result
    print('Simple interest = %.2f' %simple_interest)
    print('Total amount = %.2f' %(P + simple_interest))
```

13.

What does the if __name__ == "__main__": do in Python?

When the Python interpreter reads a source file, it executes all of the code found in it.

Before executing the code, it will define a few special variables. For example, if the python interpreter is running that module (the source file) as the main program, it sets the special __name__ variable to have a value "__main__". If this file is being imported from another module, __name__ will be set to the module's name.

One reason for doing this is that sometimes you write a module (a .py file) where it can be executed directly. Alternatively, it can also be imported and used in another module. By doing the main check, you can have that code only execute when you want to run the module as a program and not have it execute when someone just wants to import your module and call your functions themselves.

For example, if you have 2 files one.py and two.py with the following code:

one.py:

```
def func():
   print("func() in one.py")
print("Root of one.py")
if __name__ == "__main__":
   print("one.py is being run directly")
else:
   print("one.py is being imported")
```

Two.py:

```
import one
print("Root of two.py")
one.func()
if __name__ == "__main__":
   print("two.py is being run directly")
else:
   print("two.py is being imported")
```

Now if you run,

```
$ python one.py
```

You will get the output:

```
Root of one.py
one.py is being run directly
```

But if you run,

```
$ python two.py
```

You will get the output:

```
Root of in one.py
one.py is being imported
Root of in two.py
```

```
func() in one.py
two.py is being run directly
```

14.

```
# Python program to calculate compound interest using function

def compound_interest(principal, rate, time, number):
  # calculate total amount
  amount = principal * pow( 1+(rate/number), number*time)
  return amount;

# store the inputs
principal = float(input('Enter principal amount: '))
rate = float(input('Enter the interest rate: '))
time = float(input('Enter time (in years): '))
number = float(input('Enter the number of times that
                     interest is compounded per year: '))

# convert rate
rate = rate/100

# calling function
amount = compound_interest(principal, rate, time, number)
# calculate compound interest
ci = amount - principal

# display result
print('Compound interest = %.2f' %ci)
print('Total amount = %.2f' %amount)
```

15.

```
# Python Program to get the first digit of number

# take input
num = int(input('Enter any Number: '))

# get the first digit
while (num >= 10):
  num = num // 10

# printing first digit of number
print('The first digit of number:', num)
```

16.

Python program to reverse a number

```
# take inputs
num = int(input('Enter an integer number: '))

# calculate reverse of number
reverse = 0
while(num > 0):
    last_digit = num % 10
    reverse = reverse * 10 + last_digit
    num = num // 10

# display result
print('The reverse number is = ', reverse)
```

17.

```
# Python program to make a simple calculator

# take inputs
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))

# choise operation
print("Operation: +, -, *, /")
select = input("Select operations: ")

# check operations and display result
# add(+) two numbers
if select == "+":
    print(num1, "+", num2, "=", num1+num2)

# subtract(-) two numbers
elif select == "-":
    print(num1, "-", num2, "=", num1-num2)

# multiplies(*) two numbers
elif select == "*":
    print(num1, "*", num2, "=", num1*num2)

# divides(/) two numbers
elif select == "/":
    print(num1, "/", num2, "=", num1/num2)

else:
    print("Invalid input")
```

18.
```
# Python program to print fibonacci series up to n-th term

# take input
num = int(input('Enter number of terms: '))

# print fibonacci series
a, b = 0, 1
i = 0

# check if the number of terms is valid
if num <= 0:
   print('Please enter a positive integer.')

elif num == 1:
   print('The Fibonacci series: ')
   print(a)

else:
   print('The Fibonacci series: ')
   while i < num:
      print(a, end=' ')
      c = a + b
      a = b
      b = c
      i = i+1
```

19.
```
# Python program to print fibonacci series up to n-th term

# take input
num = int(input('Enter number of terms: '))

# print fibonacci series
a, b = 0, 1

# check if the number of terms is valid
if num <= 0:
   print('Please enter a positive integer.')

elif num == 1:
   print('The Fibonacci series: ')
   print(a)

else:
   print('The Fibonacci series: ')
```

```
    for i in range (1, num+1):
        print(a, end=' ')
        c = a + b
        a = b
        b = c
```

20.

```
# Python program to print multiplication table

# take inputs
num = int(input('Display multiplication table of: '))

# print multiplication table
for i in range(1, 11):
    print ("%d * %d = %d" % (num, i, num * i))
```

21.

```
# Python program to print multiplication table

# take inputs
num = int(input('Display multiplication table of: '))

# print multiplication table
i = 1
while i <= 10:
    print ("%d * %d = %d" %(num, i, num * i))
    i = i+1
```

22.

```
# Python program to find factors of a number

# take inputs
num = int(input('Enter number: '))

# find factor of number
print('The factors of', num, 'are:')
for i in range(1, num+1):
    if(num % i) == 0:
        print(i, end=' ')
```

23.

```
# Python program to find the factorial of a number

# take input
```

```
num = int(input("Enter number: "))

# check number is positive, negative, or zero
if num < 0:
  print('Factorial does not exist for negative numbers')
elif num == 0:
  print('The factorial of 0 is 1')
else:
   # find factorial of a number
  fact = 1
  for i in range(1,num + 1):
      fact = fact*i
  print('The factorial of',num,'is',fact)
```

24.

```
# Python program to convert decimal to binary

# take input
num = int(input('Enter any decimal number: '))

# display result
print('Binary value:', bin(num))
```

25.

```
# Python program to convert decimal to hexadecimal

# take inputs
num = int(input('Enter a decimal number: '))

# display result
print('HexaDecimal value = ', hex(num))
```

26.

```
# Python program to convert decimal to hexadecimal

# take inputs
num = int(input('Enter a decimal number: '))

# display result
print('HexaDecimal value = ', hex(num))
```

27.

```
# Python program to print numbers from 1 to 10

print('Numbers from 1 to 10:')
for n in range(1, 11):
```

```
   print(n, end=' ')
```

28.

```
# Python program to print numbers from 1 to 10

print('Numbers from 1 to 10:')
n = 1
while n <= 10:
   print(n, end=' ')
   n = n+1
```

Raspberry PI Programming

To get started with your Raspberry Pi computer you'll need the following accessories:

1. **A computer monitor, or television**. Most should work as a display for the Raspberry Pi, but for best results, you should use a display with HDMI input. You'll also need an appropriate display cable, to connect your monitor to your Raspberry Pi.
2. **A computer keyboard and mouse.** Any standard USB keyboard and mouse will work with your Raspberry Pi. Wireless keyboards and mice will work if already paired. For keyboard layout configuration options see raspi-config.
3. **A good quality power supply**. We recommend the official Raspberry Pi Power Supply which has been specifically designed to consistently provide +5.1V - For the Raspberry Pi 4 Model B and Raspberry Pi 400 you should use the type C power supply; for all other models you should use the micro USB power supply.
4. **Finally you'll need an SD card**; we recommend a minimum of 8GB micro SD card, and to use the Raspberry Pi Imager to install an operating system onto it. Raspberry Pi computers use a micro SD card, except for very early models which use a full-sized SD card. We recommend using an SD card of 8GB or greater capacity with Raspberry Pi OS. If you are using the lite version of Raspberry Pi OS, you can use a 4GB card.

Connecting a Display

Your Raspberry Pi has an HDMI port which you can connect directly to a monitor or TV with an HDMI cable. This is the easiest solution; some modern monitors and TVs have HDMI ports, some do not, but there are other options.

NOTE

The Raspberry Pi 4 has two micro HDMI connectors, which require a good-quality micro HDMI cable, especially when using 4K monitors or television. Raspberry Pi sells a suitable cable. If you're using your Raspberry Pi with a monitor with built-in speakers and are connecting to it using an HDMI cable you can also use it to output sound.

Installing Raspberry Pi OS:

Raspberry Pi Imager is the quick and easy way to install an operating system to a microSD card ready to use with your Raspberry Pi.

Following images are available on official website of Pi

- Raspberry Pi OS
- Raspberry Pi OS (64-bit)
- Raspberry Pi OS (Legacy)
- Raspberry Pi Desktop

To use PI use following commands to use library.
1.pip install RPi.GPIO
2. pip install gpiozero

GPIO pins
GPIO stands for General-Purpose Input/Output. These pins are a physical interface between the Raspberry Pi and the outside world.

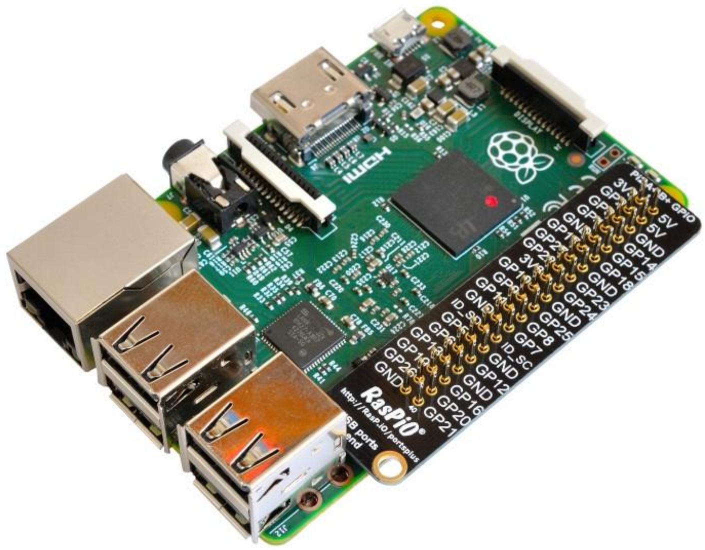
At the simplest level, you can think of them as switches that you can turn on or off (input) or that the Pi can turn on or off (output).

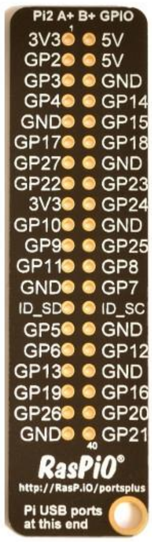
The GPIO pins allow the Raspberry Pi to control and monitor the outside world by being connected to electronic circuits.

The Pi is able to control LEDs, turning them on or off, run motors, and many other things. It's also able to detect whether a switch has been pressed, the temperature, and light. We refer to this as physical computing.

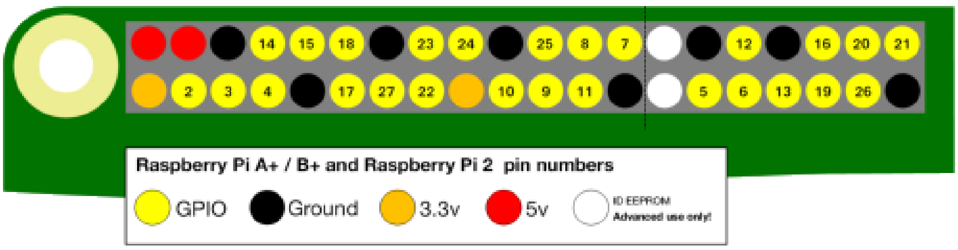
There are 40 pins on the Raspberry Pi (26 pins on early models), and they provide various different functions.

If you have a RasPiO pin label, it can help to identify what each pin is used for. Make sure your pin label is placed with the keyring hole facing the USB ports, pointed outwards.





If you don't have a pin label, then this guide can help you to identify the pin numbers:

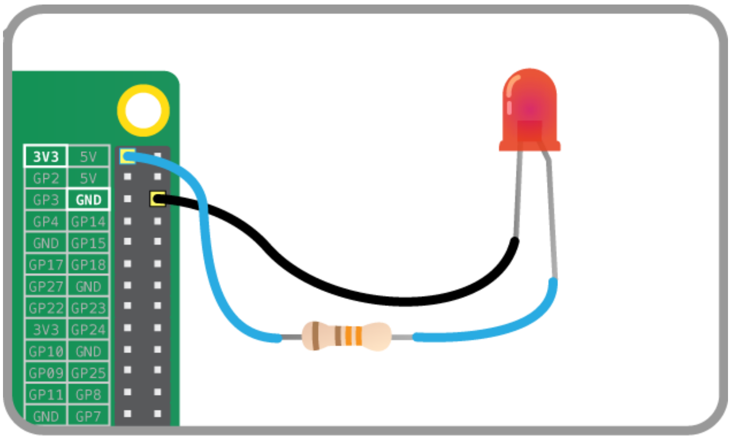


You'll see pins labelled as 3V3, 5V, GND and GP2, GP3, etc:

Interface LED with Raspberry PI

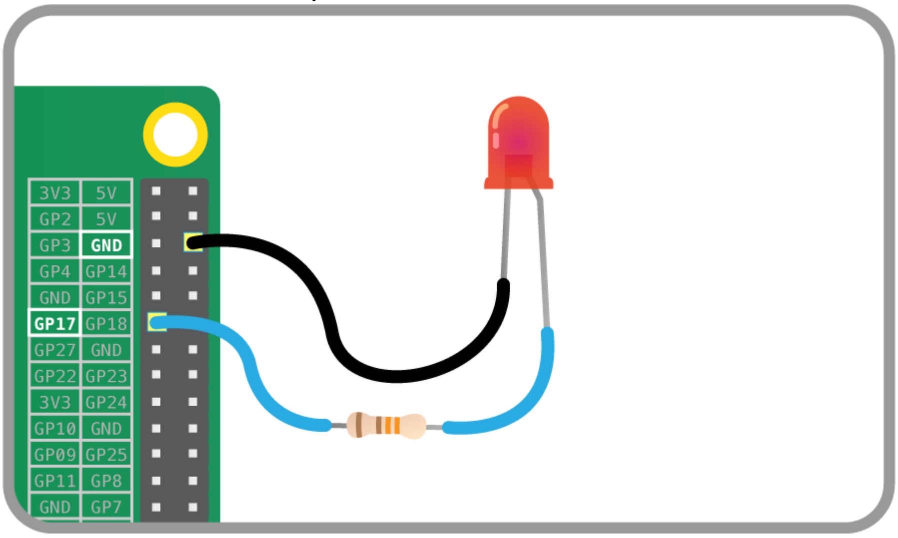
LEDs are delicate little things. If you put too much current through them they will pop (sometimes quite spectacularly). To limit the current going through the LED, you should always use a resistor in series with it.

Try connecting the long leg of an LED to the Pi's 3V3 and the short leg to a GND pin. The resistor can be anything over about 50Ω.



The LED should light up. It will always be on, because it's connected to a 3V3 pin, which is itself always on.

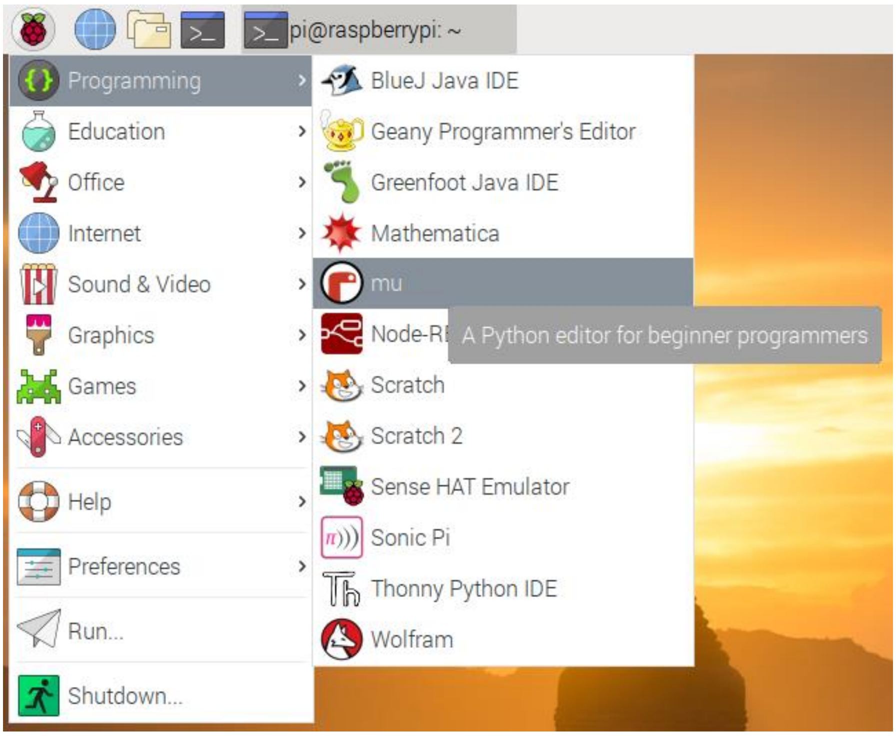
Now try moving it from 3V3 to GPIO pin 17:



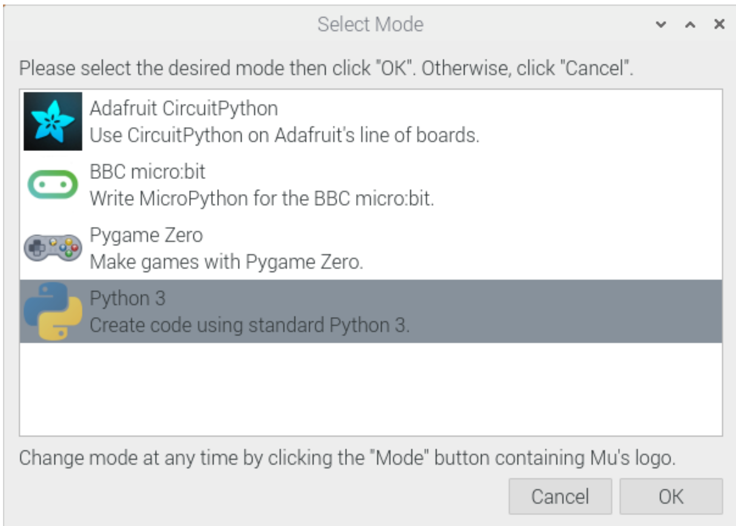
The LED should now turn off, but now it's on a GPIO pin, and can therefore be controlled by code.

GPIO Zero is a new Python library which provides a simple interface to everyday GPIO components. It comes installed by default in Raspbian.

How to open Mu (Python editor)



Then choose the mode in which you want to use Mu. Choose Python 3 if you are creating a new Python script.



With the help of the time library and a little loop, you can make the LED flash.

1. Create a new file by clicking New.
2. Save the new file by clicking Save. Save the file as gpio_led.py.
3. Enter the following code to get started:
4. from gpiozero import LED
5. Enter the following code to get sarted.

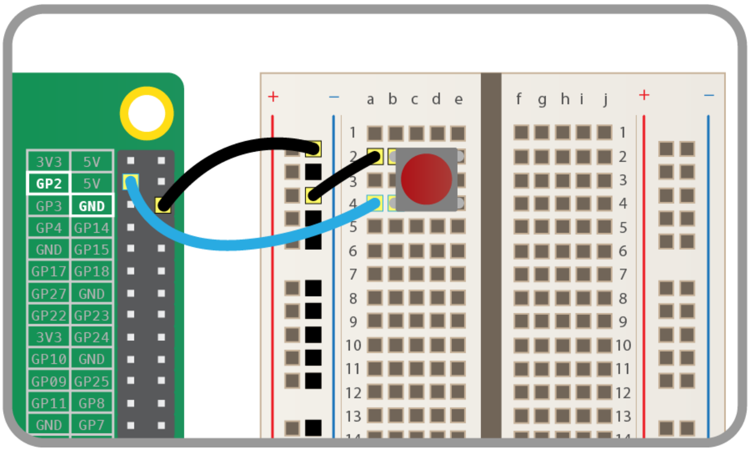
```
from gpiozero import LED→it is used to import LED object to program
from time import sleep→It is used to bring sleep object to program
led = LED(17)→attached led to pin 17
while True:→create infinite loop
   led.on()→make led ON
   sleep(1)→sleep(delay) for 1 second
   led.off()→make led OFF
  sleep(1)→sleep (delay) for 1 second
```

```
led.on()
while True:
    led.toggle()
    sleep(1)
```

6. Save the file and run the code with by clicking on Run.
7. The LED should be flashing on and off. To exit the program click Stop.

Using buttons to get input

Now you're able to control an output component (an LED), let's connect and control an input component: a button. Connect a button to another GND pin and GPIO pin 2, like this:



Create a new file by clicking New. Save the new file by clicking Save. Save the file as gpio_button.py.

Code:

from gpiozero import LED, Button

```
from time import sleep

led = LED(17)
button = Button(2)

button.wait_for_press()
led.on()
sleep(3)
led.off()
```

Making a switch

With a switch, a single press and release on the button would turn the LED on, and another press and release would turn it off again. Modify your code so that it looks like this:

```
from gpiozero import LED, Button
from time import sleep

led = LED(17)
button = Button(2)
while True:
    button.wait_for_press()
    led.toggle()
    sleep(0.5)
```

led.toggle() switches the state of the LED from on to off, or off to on.

Since this happens in a loop the LED will turn on and off each time the button is pressed.

It would be great if you could make the LED switch on only when the button is being held down. With GPIO Zero, that's easy. There are two methods of the Button class called when_pressed and when_released. These don't block the flow of the program, so if they are placed in a loop, the program will continue to cycle indefinitely.

Modify your code to look like this:

```
from gpiozero import LED, Button
from signal import pause
```

```
led = LED(17)
button = Button(2)

button.when_pressed = led.on
button.when_released = led.off

pause()
```

Calling pause() is required to keep the program listening for the different events. If this wasn't present, then the program would run once and exit.

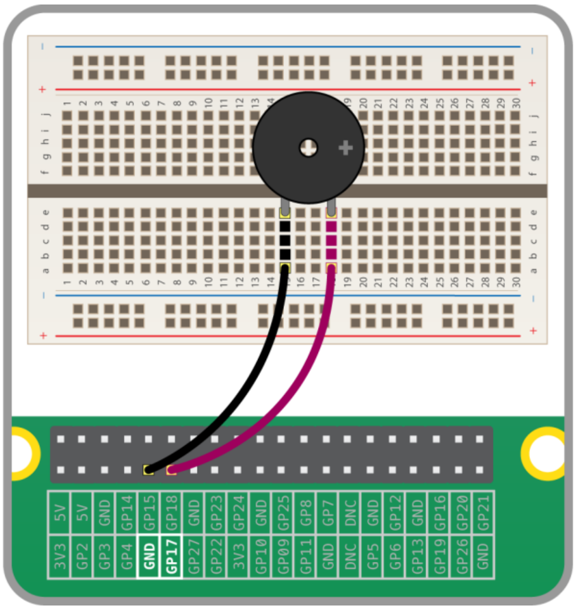
Using a buzzer

There are two main types of buzzer: *active* and *passive*.

A *passive* buzzer emits a tone when a voltage is applied across it. It also requires a specific signal to generate a variety of tones. The *active* buzzers are a lot simpler to use, so these are covered here.

Connecting a buzzer

An *active* buzzer can be connected just like an LED, but as they are a little more robust, you won't be needing a resistor to protect them.
Set up the circuit as shown below:



Add Buzzer to the from gpiozero import... line:

Code:

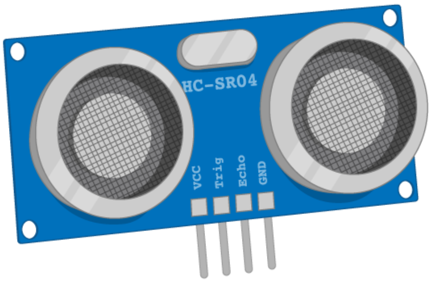
```
from gpiozero import Buzzer
from time import sleep

buzzer = Buzzer(17)

while True:
   buzzer.on()
   sleep(1)
   buzzer.off()
   sleep(1)
```

A Buzzer has a beep() method which works like an LED’s blink. Try it out:

```
while True:
   buzzer.beep()
```

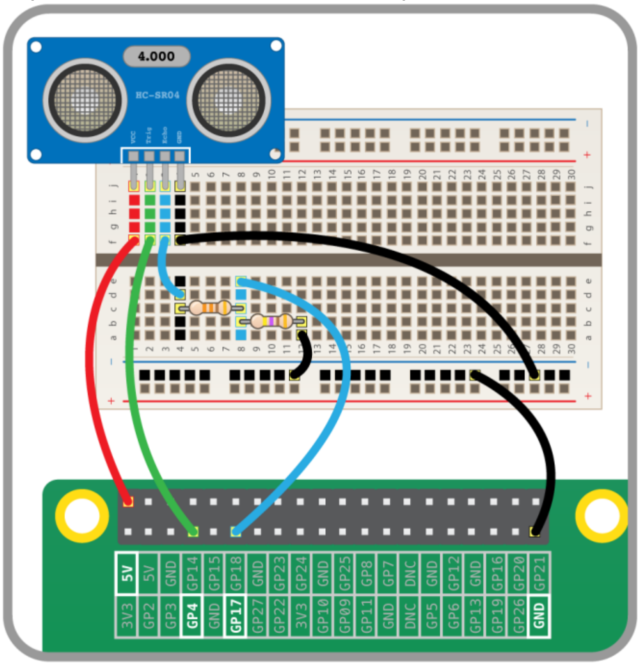
Using an ultrasonic distance sensor

In air, sound travels at a speed of 343 metres per second. An ultrasonic distance sensor sends out pulses of ultrasound which are inaudible to humans, and detects the echo that is sent back when the sound bounces off a nearby object. It then uses the speed of sound to calculate the distance from the object.



Wiring

The circuit connects to two GPIO pins (one for echo, one for trigger), the ground pin, and a 5V pin. You'll need to use a pair of resistors (330Ω and 470Ω) as a potential divider:



Code

To use the ultrasonic distance sensor in Python, you need to know which GPIO pins the echo and trigger are connected to.

Use the following code to read the DistanceSensor.

```
from gpiozero import DistanceSensor
ultrasonic = DistanceSensor(echo=17, trigger=4)
```

```
    while True:
        distance = ultrasonic.distance
        print("distance")
        print(distance)
        //print(ultrasonic.distance)
```

The value should get smaller the closer your hand is to the sensor.

Use a loop to print different messages when the sensor is in range or out of range:

```
from gpiozero import DistanceSensor
ultrasonic = DistanceSensor(echo=17, trigger=4, threshold_distance=0.5)
while True:
   ultrasonic.wait_for_in_range()
   print("In range")
   ultrasonic.wait_for_out_of_range()
   print("Out of range")
```

Now wave your hand in front of the sensor; it should switch between showing the message "In range" and "Out of range" as your hand gets closer and further away from the sensor.

The default range threshold is 0.3m. This can be configured when the sensor is initiated:

ultrasonic = DistanceSensor(echo=17, trigger=4, threshold_distance=0.5)

Alternatively, this can be changed after the sensor is created, by setting the threshold_distance property:
ultrasonic.threshold_distance = 0.5

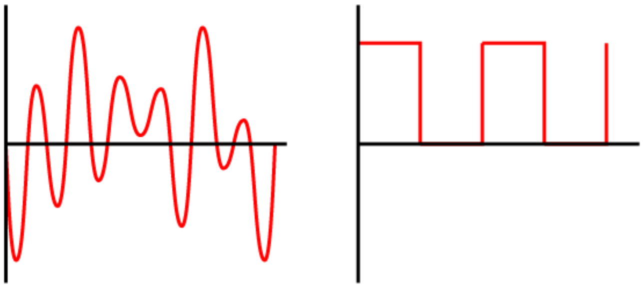
The wait_for functions are blocking, which means they halt the program until they are triggered. Another way of doing something when the sensor goes in and out of range is to use when properties, which can be used to trigger actions in the background while other things are happening in the code.

<u>Using a light-dependent resistor (LDR)</u>

In the world of engineering, there are two type of input and output (I/O): analogue and digital. Digital I/O is fairly easy to understand; it's either *on* or *off*, *1* or *0*.

When talking about voltages and the Raspberry Pi, any input that is approximately below 1.8V is considered *off* and anything above 1.8V is considered *on*. For output, 0V is off and 3.3V is on.

Analogue I/O is a little trickier. With an analogue input, we can have a range of voltages from 0V up to 3.3V, and the Raspberry Pi is unable to detect exactly what that voltage is.



Analogue inputs could be given a range of voltages, anywhere from 0V up to 3.3V.

Digital inputs are either on or off. Any voltage above 1.8V is considered on, and below that is considered off.

How, then, can we use a Raspberry Pi to determine the value of an analogue input, if it can only tell when the voltage to a GPIO pin goes above 1.8V?

Using a capacitor for analogue inputs

Capacitors are electrical components that store charge.

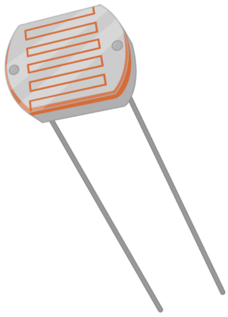
When current is fed into a capacitor, it will begin to store charge. The voltage across the capacitor will start off low, and increase as the charge builds up.

If current flow is stopped then it will start discharging. If a load is connected across capacitor then it will start feeding voltage to that load and discharge.

By putting a resistor in series with the capacitor, you can slow the speed at which it charges. With a high resistance, the capacitor will charge slowly, whereas a low resistance will let it charge quickly.

Light-dependent resistors

An LDR (sometimes called a photocell) is a special type of resistor.



When light hits the LDR, its resistance is very low, but when it is in the dark its resistance is very high.

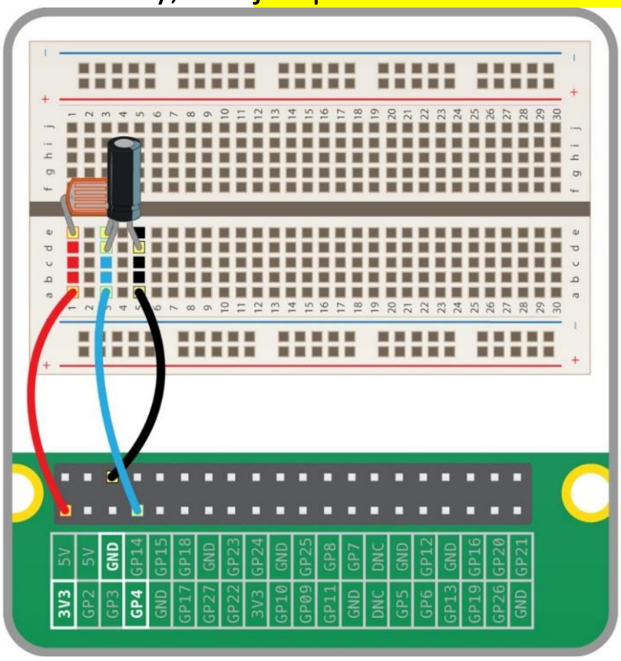
By placing a capacitor in series with an LDR, the capacitor will charge at different speeds depending on whether it's light or dark.

Que: Creating a light-sensing circuit / Interface LDR with raspberry PI.

- Place an LDR into your breadboard.

- Now place a capacitor in series with the LDR. As the capacitor is a polar component, you must make sure the positive, long leg is on the same track as the LDR leg.

- Finally, add jumper leads to connect the two components to your Raspberry Pi.



<u>Coding a light sensor</u>

Use the following code to set up the light sensor:

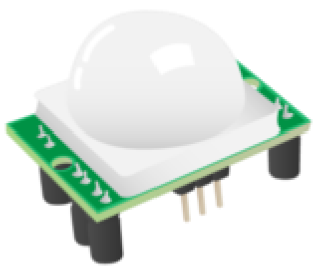
```
from gpiozero import LightSensor, Buzzer

ldr = LightSensor(4)  # alter if using a different pin
while True:
   print(ldr.value)
   if ldr.value>30:
         buzzer.on()
         sleep(2)
         buzzer.off()
```

Run this code, then cover the LDR with your hand and watch the value change. Try shining a strong light onto the LDR.

Using a PIR sensor

- Humans and other animals emit radiation all the time.
- This is nothing to be concerned about, though, as the type of radiation we emit is infrared radiation (IR), which is pretty harmless at the levels at which it is emitted by humans.
- In fact, all objects at temperatures above absolute zero (-273.15C) emit infrared radiation.

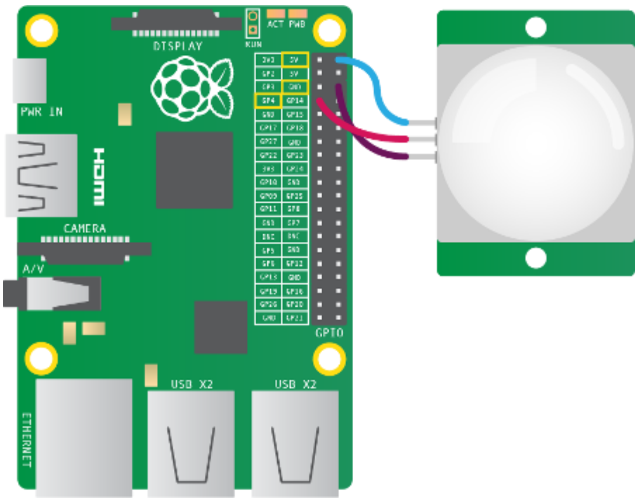
- A PIR sensor detects changes in the amount of infrared radiation it receives.
- When there is a significant change in the amount of infrared radiation it detects, then a pulse is triggered.
- This means that a PIR sensor can detect when a human (or any animal) moves in front of it.



Wiring a PIR sensor

- The pulse emitted when a PIR detects motion needs to be amplified, and so it needs to be powered.
- **There are three pins on the PIR: they should be labelled Vcc, Gnd, and Out.**

- These labels are sometimes concealed beneath the Fresnel lens (the white cap), which you can temporarily remove to see the pin labels.



1. **As shown above, the Vcc pin needs to be attached to a 5V pin on the Raspberry Pi.**
2. **The Gnd pin on the PIR sensor can be attached to any ground pin on the Raspberry Pi.**
3. **Lastly, the Out pin needs to be connected to any of the GPIO pins (4).**

Detecting motion

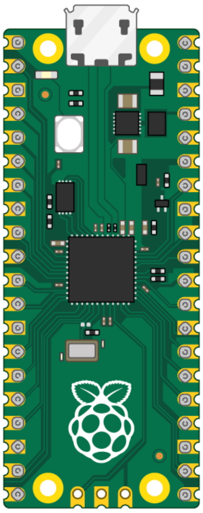
- You can detect motion with the PIR using the code below:

```
from gpiozero import MotionSensor

pir = MotionSensor(4)

while True:
    pir.wait_for_motion() →it will wait for any kind of motion
    print("You moved")→as soon as motion is detected it print this message
    pir.wait_for_no_motion()→wait for motion to be stopped.
```

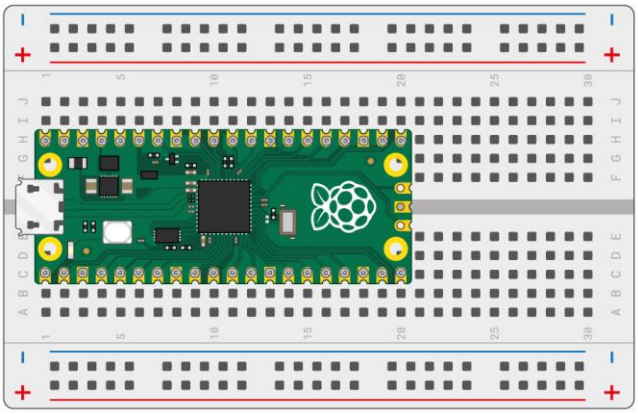
Raspberry PI PICO:

Meet Raspberry Pi Pico

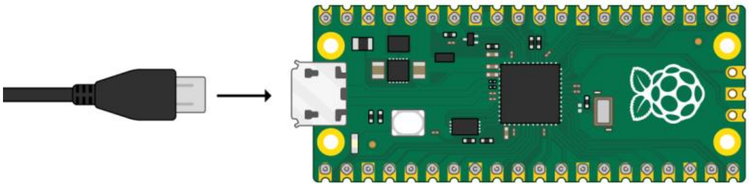
This is a Raspberry Pi Pico. Hopefully your device has already had the header pins soldered on.



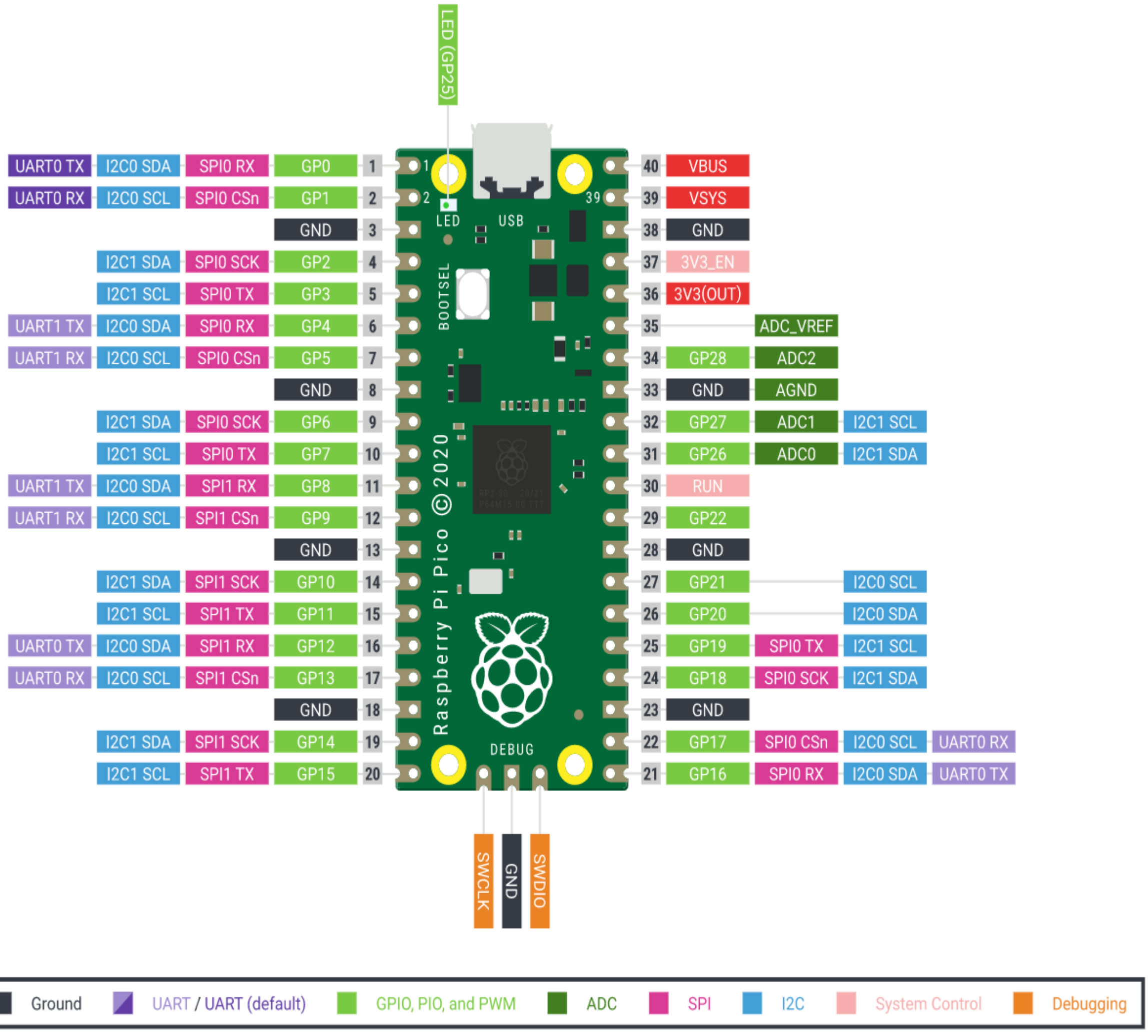
If you have a breadboard, put your Raspberry Pi Pico on the board. Place it so that the two headers are separated from the middle.



Plug your micro USB cable into the port on the left-hand side of the board.



If you need to know the pin numbers for a Raspberry Pi Pico, you can refer to the following diagram.



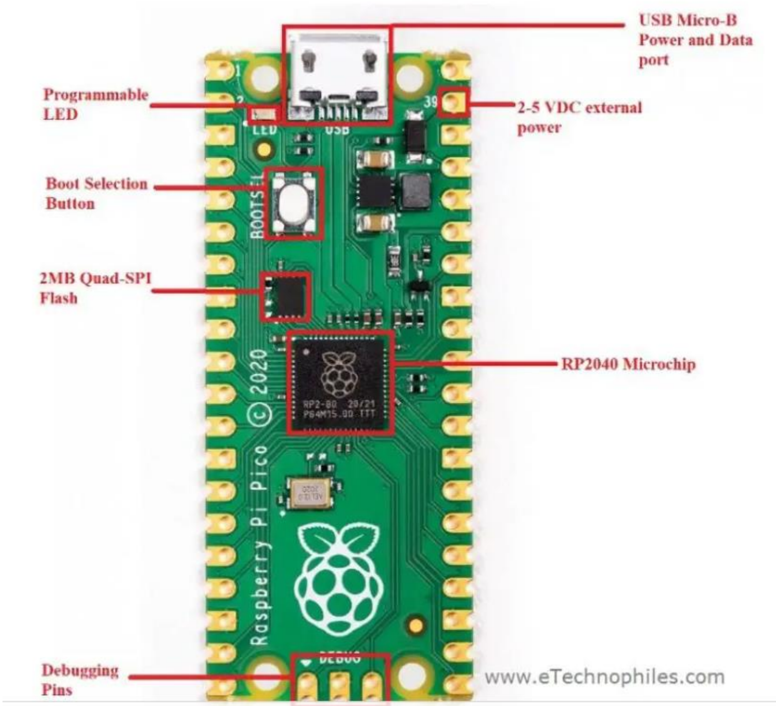
Raspberry Pi PICO Board Layout

Raspberry Pi Pico is made up of several components. The board layout given above shows some of them: **RP2040 Microcontroller, Debugging pins, Flash Memory, Boot selection button, programmable LED, USB port, and power pin.**

What is RP2040 Microcontroller Chip?

RP2040 microcontroller is a custom-designed processor chip by the Raspberry Pi foundation itself. It is a powerful but cost-effective processor, featuring a dual-core Arm Cortex-M0+ processor running at 133Mhz.

It has 264KB of internal RAM and support for up to 16MB of on board flash memory.



It has 264KB of internal RAM and support for up to 16MB of onboard flash memory. A wide range of flexible I/O operations is possible including I2C, SPI, and Programmable general purpose I/O (GPIO).

What is the meaning of RP2040?

The name of the RP2040 microcontroller is made up of 5 sections:

- RP in RP2040 means 'Raspberry Pi'.
- 2 signifies the number of processor cores the microcontroller has i.e, 2 cores.
- 0 represents the type of processor core the RP2040 microcontroller has. This processor is called ARM Cortex-M0+ processor. Other microcontrollers of this series by ARM are named as Cortex-M1, Cortex-M3, Cortex-M4, Cortex-M7 etc.
- 4 means the microcontroller has has 264 kilobytes (kB) of RAM. Which is based on a special mathematical function: floor(log2(RAM/16)).
- 0 simply means there is no non-volatile storage on-board.

The Raspberry Pi Pico pinout shows that it has a total of 40 pins including GND and Vcc pins. The pins can be categorized as Power, ground, UART, GPIO, PWM, ADC, SPI, I2C, system control, and Debugging pins.

Unlike the raspberry pi computer board series, GPIO pins present on the Pico board have multiple functions.

For example, GP4 and GP5 pins can be either used as digital input or digital output or I2C1 (SDA and SCK pins) or UART1 (Rx and Tx). But only one function can be enabled at a time.

How many GPIO Pins are there on Pi Pico?

In Raspberry Pi Pico, out of 40 pins, 26 pins are multi-functional GPIO pins. These 26 GPIO pins can be used both as digital input and digital output. These digital pins are marked from GP0, GP1, and up to GP28.

The marked GP23, GP24, GP25, and GP29 are not exposed on the header. These GPIO pins are used for internal board functions:

How many Analog Pins are in Pi Pico?

The board comes with 4 analog pins with 12-bit ADC. Hence, we can use these pins to read analog inputs from various sensors.

But one out of these four pins(ADC 4) is not provided as a GPIO pin on the board. This fourth ADC pin is internally connected to a temperature sensor.

NOTE: Therefore, to measure the temperature we can directly use this build-in temperature by reading the analog value of ADC4.

PWM Pins on R-Pi Pico

The Raspberry Pi Pico has 8 PWM blocks/slices(1-8) and each PWM block provides up to two PWM outputs(A-B). This means each block can drive up to two PWM outputs.

Hence, there is a total of 16 PWM output channels available at a time and every GPIO pin on the pico is capable of generating PWM output.

So all GPIO pins can be configured to get PWM signal output when required but two GPIO pins with the same PWM designation cannot be used at once.

What is the PIO feature in Pi Pico?

Pico's Programmable Input/Output (PIO) feature lets you configure the input and output pins on Pico to suit your needs. This is done through a software interface, which means that no hardware changes are necessary.

The PIO can be used to control external devices, like sensors or motors. It can also be used to interface with other digital systems. The PIO is a powerful and versatile tool that lets users have a lot of control over their Pico's behavior.

Each RP2040 has two PIO chips, and each chip can do things like small processors. They can do this by using instructions that are stored in shared PIO memory.

We can utilize the PIO instances to mimic more sophisticated peripherals that aren't directly supported by the chip, such as the SD card interface, CAN Bus, and WS2812b driver. The PIO state machines run independently of the main CPU, allowing these simulated peripherals to communicate with external devices simultaneously with the primary program.

I2C Pins on Raspberry Pi Pico

I2C is a two-wire, bi-directional serial bus that provides an easy and quick method for transmission of transmission over a short distance between I2C enabled devices.

The Raspberry Pi Pico comes with two I2C controllers, both I2C controllers are accessible through GPIO pins of Raspberry Pi Pico.

I2C Controller	**GPIO Pins**
I2C0 SDA	GP0/GP4/GP8/GP12/GP16/GP20
I2C0 SCL	GP1/GP5/GP9/GP13/GP17/GP21
I2C1 SDA	GP2/GP6/GP10/GP14/GP18/GP26
I2C1 SCL	GP3/GP7/GP11/GP15/GP19/GP27

This Table shows the I2C pins on Pi Pico

SPI Pins on Raspberry Pi Pico

Serial Peripheral Interface (**SPI**) is an interface bus that is used to transfer data between the microcontroller and SPI-enabled devices. Raspberry Pi Pico supports two SPI interfaces that are accessible through GPIO pins of the board.

SPI Controller	GPIO Pins
SPI0_RX	GP0/GP4/GP16
SPI0_TX	GP3/GP7/GP19
SPI0_CLK	GP2/GP6/GP18
SPI0_CSn	GP1/GP5/GP17
SPI1_RX	GP8/GP12
SPI1_TX	GP11/GP15
SPI1_CLK	GP10/GP14
SPI1_CSn	GP9/GP13

This Table shows the SPI pins on Pi Pico

UART Pins on Pico

The Raspberry Pi Pico also contains two identical UART peripherals. UART (universal asynchronous receiver-transmitter) pins are used for asynchronous serial communication between the micro-controller and UART devices or other microcontrollers.

UART Pins	GPIO Pins
UART0-TX	GP0/GP12/GP16
UART0-RX	GP1/GP13/GP17
UART1-TX	GP4/GP8
UART1-RX	GP5/GP9

UART Pins on Pico

GPIO Interrupts on Pico

All GPIO pins on the board can be configured as an external interrupt if one of the following changes in the state of GPIO pins occur:

1. Level high(+3v)
2. Level Low(GND)
3. Positive edge (transition from active low to active high)
4. Negative edge (transition from active high to active low)

Other Pins on Pico board:-

GND: is the Ground pin used to complete the circuit.

VBUS: is the micro-USB input voltage it is connected to micro-USB port pin 1.

VSYS: is the main system input voltage, which can vary in the allowed range 1.8V to 5.5V, and is used by the on-board

SMPS: to generate the 3.3V for the board and its GPIO.

3V3_EN: connects to the onboard SMPS enable pin, and is set to high (to VSYS) connected through a 100K resistor. To disable the 3.3V(which also de-powers the RP2040), short this pin low.

3V3: is the main 3.3V supply to the board and its I/O, generated by the on-board SMPS. This pin can also be used to power external components.

ADC_VREF: is the ADC pin power supply voltage, and is generated by filtering the 3.3V supply coming through the board.

AGND: is the ground reference for GPIO26-29, there is a separate analog ground plane running under these signals and terminating at this pin. If the ADC pins are not used or ADC performance is doesn't matter, this pin can be connected to digital ground.

RUN: is the RP2040 enable pin, and has an internal (on-chip) pull-up resistor to 3.3V of about ~50K Ohms. To resetRP2040, short this pin low.

Specifications of R-PI Pico:

- Dual-core Arm Cortex M0+ processor running at 133 MHz
- Board comes with 264KB of SRAM, and 2MB of onboard Flash memory
- Comes with the castellated module that allows us to solder it directly to carrier boards
- USB 1.1 with device and host support
- Low-power sleep and dormant modes
- Drag-and-drop programming using mass storage over USB
- 26 × multi-function GPIO pins
- 2 × SPI peripherals, 2 × I2C controllers, 2 × UART peripherals, 3 × 12-bit ADC, 16 × controllable PWM channels
- Accurate clock and timer on-chip
- Temperature sensor

- Accelerated floating-point libraries on-chip
- It features 8 Programmable I/O (PIO) state machines for custom peripheral support.